

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T09:58:20

Source: data/json/noise_experiments/v2/noise2i_cap6_game/claude-sonnet-4.5/game/noisy_bidirectional_informed_memprimary_cap6/noise2i_cap6_game_004_neutral_rep04.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary_cap6
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 4
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 1
coplayer_history_window: 0
show_agent_names: true
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 4
task_order: [
  "game"
]
agent_names: {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
}
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

ROUND 1

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$3; payoffs Agent_1 (Aster)=\$5, Agent_2 (Briar)=\$6; noise/visible: sent visible=\$2.21, received visible=\$6.63, returned visible=\$2.94

Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 9.0, "min": 0, "raw": 3.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 5.0,

"Agent_2": 6.0

}

Visible balances after round: {

"Agent_1": 4.9399999999999995,

"Agent_2": 3.63

```
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:55:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance the potential for mutual gain against the risk of the receiver not returning anything.

Sending money creates value since it's tripled, but requires trust. Since this is round 1 with no history, I'll take a moderate approach that signals cooperation while managing risk.

I'll send **\$3**:

- This gets tripled to \$9 for the receiver
- I initially keep \$2
- If the receiver reciprocates reasonably, we both benefit
- If not, I haven't lost everything

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- Sending creates mutual benefit (money is tripled)
- But I'm vulnerable to the receiver keeping everything
- This is round 1, so I have no history with this receiver
- The note mentions there may be "communication noise" - amounts may differ from what was actually sent/returned

Strategy options:

1. Send \$0 - Safe but no value creation, I keep \$5
2. Send some amount - Creates value but requires trust
3. Send \$5 - Maximum value creation but maximum risk

In experimental trust games, common first-round strategies include:

- Sending about half (\$2-3) as a moderate trust signal
- This allows for value creation while limiting downside risk
- If the receiver is reciprocal, both players can benefit

Given this is round 1 and I have no information about the receiver, I'll send a moderate amount to signal willingness to cooperate while not being completely vulnerable.

I'll send \$3:

- I keep \$2 initially
- Receiver gets \$9 (3 × 3)
- If receiver returns even \$2-3, we both benefit compared to sending nothing
- This signals trust without being naive

Let me format this properly as JSON.

```
Usage: {
  "input_tokens": 365,
  "output_tokens": 472,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:55:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
```

```

"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.21, that is 44.2% of their total endowment. You received \$6.63. How much do you return to The sender? (0-6.63)

Assistant output:

As the RECEIVER, I need to decide how much to return to the sender who sent me \$2.21 (44.2% of their endowment).

The sender showed moderate trust by sending nearly half their endowment. I received \$6.63 after the 3x multiplier.

Let me consider a fair return:

- The sender currently has: $\$5 - \$2.21 = \$2.79$
- If I return $\$3.00$, the sender ends up with $\$5.79$ (gaining $\$0.79$ from the exchange)
- I would keep $\$3.63$ (gaining $\$3.63$ from the exchange)

This seems like a reasonable split that rewards the sender's trust while both of us benefit from cooperation.

```
{'return': 3.00}
```

Reasoning:

Let me think through this situation carefully.

As the RECEIVER in Round 1:

- The sender sent $\$2.21$ out of their $\$5$ endowment (44.2%)
- I received $\$6.63$ (which is $\$2.21 \times 3$)
- I need to decide how much to return (between $\$0$ and $\$6.63$)

Key considerations:

1. The sender showed moderate trust by sending 44.2% of their endowment
2. If I return nothing, I keep $\$6.63$ and the sender gets only $\$2.79$ ($\$5 - \2.21)
3. If I return a fair amount, we can both benefit

The sender currently has: $\$5 - \$2.21 = \$2.79$

I currently have: $\$6.63$

A common approach in trust games is to:

- Return enough so the sender ends up with more than their initial $\$5$
- But also keep some profit for myself

If I return $\$3.00$:

- Sender ends up with: $\$2.79 + \$3.00 = \$5.79$ (a gain of $\$0.79$)
- I keep: $\$6.63 - \$3.00 = \$3.63$ (a gain of $\$3.63$)

If I return $\$2.50$:

- Sender ends up with: $\$2.79 + \$2.50 = \$5.29$ (a gain of $\$0.29$)
- I keep: $\$6.63 - \$2.50 = \$4.13$ (a gain of $\$4.13$)

If I return $\$3.50$:

- Sender ends up with: $\$2.79 + \$3.50 = \$6.29$ (a gain of $\$1.29$)
- I keep: $\$6.63 - \$3.50 = \$3.13$ (a gain of $\$3.13$)

...

```
Usage: {
  "input_tokens": 394,
  "output_tokens": 711,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent $\$2.5$; received $\$7.5$; Agent_1 (Aster) returned $\$0$; payoffs Agent_2 (Briar)=\$ 2.5 , Agent_1 (Aster)=\$ 7.5 ; noise/visible: sent visible=\$ 0 , received visible=\$ 0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 7.5, "min": 0, "raw": 0.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent_1": 12.5,

"Agent_2": 8.5

}

Visible balances after round: {

"Agent_1": 4.9399999999999995,

"Agent_2": 6.13

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:55:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.21, that is 44.2% of their total endowment. You received \$6.63. How much do you return to The sender? (0-6.63)

Message 3 role=assistant

content:

As the RECEIVER, I need to decide how much to return to the sender who sent me \$2.21 (44.2% of their endowment).

The sender showed moderate trust by sending nearly half their endowment. I received \$6.63 after the 3x multiplier.

Let me consider a fair return:

- The sender currently has: $\$5 - \$2.21 = \$2.79$
- If I return \$3.00, the sender ends up with \$5.79 (gaining \$0.79 from the exchange)
- I would keep \$3.63 (gaining \$3.63 from the exchange)

This seems like a reasonable split that rewards the sender's trust while both of us benefit from cooperation.

```
{'return': 3.00}
```

reasoning:

Let me think through this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$2.21 out of their \$5 endowment (44.2%)
- I received \$6.63 (which is $\$2.21 \times 3$)
- I need to decide how much to return (between \$0 and \$6.63)

Key considerations:

1. The sender showed moderate trust by sending 44.2% of their endowment
2. If I return nothing, I keep \$6.63 and the sender gets only \$2.79 ($\$5 - \2.21)
3. If I return a fair amount, we can both benefit

The sender currently has: $\$5 - \$2.21 = \$2.79$

I currently have: \$6.63

A common approach in trust games is to:

- Return enough so the sender ends up with more than their initial \$5
- But also keep some profit for myself

If I return \$3.00:

- Sender ends up with: $\$2.79 + \$3.00 = \$5.79$ (a gain of \$0.79)
- I keep: $\$6.63 - \$3.00 = \$3.63$ (a gain of \$3.63)

If I return \$2.50:

- Sender ends up with: $\$2.79 + \$2.50 = \$5.29$ (a gain of \$0.29)
- I keep: $\$6.63 - \$2.50 = \$4.13$ (a gain of \$4.13)

If I return \$3.50:

- Sender ends up with: $\$2.79 + \$3.50 = \$6.29$ (a gain of \$1.29)
- I keep: $\$6.63 - \$3.50 = \$3.13$ (a gain of \$3.13)

...

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 711,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.63.

How much do you send to the receiver this round? (0-5)

Assistant output:

As the SENDER in Round 2, I need to decide how much to send.

In Round 1, the sender showed trust by sending me 44.2% of their endowment, and I reciprocated by returning a fair amount. This established a cooperative pattern.

Following this cooperative approach, I'll send a similar proportion to continue building mutual

l trust.

```
{'send': 2.50}
```

Reasoning:

Now I'm the SENDER in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$2.21 (44.2% of their \$5)
- I received \$6.63
- I returned \$3.00
- The sender ended with $\$5 - \$2.21 + \$3.00 = \5.79
- I kept \$3.63

Now in Round 2, I'm the sender. I need to think about:

1. The previous sender showed moderate trust by sending 44.2%
2. I reciprocated fairly by returning about 45% of what I received
3. This established a cooperative pattern

Since the previous sender sent about 44% of their endowment and I returned a fair amount, I should consider sending a similar or slightly higher amount to continue building trust and cooperation.

Sending around \$2-2.50 seems reasonable:

- Shows trust and willingness to cooperate
- Follows the pattern established in Round 1
- Gives the receiver a meaningful amount to work with (\$6-\$7.50 after multiplier)

I'll send \$2.50, which is 50% of my endowment - slightly more than what was sent to me in Round 1, showing I'm willing to continue cooperative behavior.

Usage: {

```
"input_tokens": 617,  
"output_tokens": 399,  
"reasoning_tokens": 0
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:55:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_1",  
"move_index": 1,  
"opponent_id": "Agent_2",  
"pairing": {  
  "agent_names": {  
    "Agent_1": "Aster",  
    "Agent_2": "Briar"  
  },  
  "agents": [  
    "Agent_2",  
    "Agent_1"  
  ],  
  "investor": "Agent_2",  
  "roles": {  
    "Agent_1": "trustee",  
    "Agent_2": "investor"  
  },  
  "trustee": "Agent_1"  
},  
"role": "trustee",  
"role_label": "Receiver",  
"round": 2,  
"task": "game",  
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance the potential for mutual gain against the risk of the receiver not returning anything.

Sending money creates value since it's tripled, but requires trust. Since this is round 1 with no history, I'll take a moderate approach that signals cooperation while managing risk.

I'll send **\$3**:

- This gets tripled to \$9 for the receiver
- I initially keep \$2
- If the receiver reciprocates reasonably, we both benefit
- If not, I haven't lost everything

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- Sending creates mutual benefit (money is tripled)
- But I'm vulnerable to the receiver keeping everything
- This is round 1, so I have no history with this receiver
- The note mentions there may be "communication noise" - amounts may differ from what was actually sent/returned

Strategy options:

1. Send \$0 - Safe but no value creation, I keep \$5
2. Send some amount - Creates value but requires trust
3. Send \$5 - Maximum value creation but maximum risk

In experimental trust games, common first-round strategies include:

- Sending about half (\$2-3) as a moderate trust signal
- This allows for value creation while limiting downside risk
- If the receiver is reciprocal, both players can benefit

Given this is round 1 and I have no information about the receiver, I'll send a moderate amount to signal willingness to cooperate while not being completely vulnerable.

I'll send \$3:

- I keep \$2 initially
- Receiver gets \$9 (3×3)
- If receiver returns even \$2-3, we both benefit compared to sending nothing
- This signals trust without being naive

Let me format this properly as JSON.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 472,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.9399999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent me \$0, I received \$0 (after the 3x multiplier).

I have no money to return, so I must return **\$0**.

```
'''json
{"return": 0}
'''
```

Reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 2
- The sender sent me \$0
- Since it's multiplied by 3x, I received \$0 ($0 \times 3 = 0$)
- I can only return between \$0 and \$0 (what I received)
- My total earnings across all rounds are \$4.94

Wait, let me think about Round 1:

- I was the SENDER and sent \$3
- That would be tripled to \$9 for the receiver
- The receiver apparently returned some amount
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$4.94
- So I must have received back: $\$4.94 - \$2 = \$2.94$

That makes sense - the receiver got \$9 and returned about \$2.94 to me, keeping about \$6.06. That's roughly 1/3 back, which is a reasonable but somewhat low reciprocation.

Now in Round 2:

- The sender sent me \$0
- So I have \$0 to work with
- I can only return \$0

The instructions mention there may be communication noise, but in this case, even with noise, if the sender sent close to \$0, there's very little I can return.

I need to return \$0 since that's all I have.

```
Usage: {
  "input_tokens": 561,
  "output_tokens": 377,
  "reasoning_tokens": 0
}
```

ROUND 3

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$2, Agent_2 (Briar)=\$9; noise/visible: sent visible=\$0, received visible=\$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 9.0, "min": 0, "raw": 0.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 14.5,
"Agent_2": 17.5

}

Visible balances after round: {

"Agent_1": 6.9399999999999995,
"Agent_2": 6.13

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:55:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"

},

"agents": [
 "Agent_1",
 "Agent_2"

],

"investor": "Agent_1",

"roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance the potential for mutual gain against the risk of the receiver not returning anything.

Sending money creates value since it's tripled, but requires trust. Since this is round 1 with no history, I'll take a moderate approach that signals cooperation while managing risk.

I'll send **\$3**:

- This gets tripled to \$9 for the receiver
- I initially keep \$2
- If the receiver reciprocates reasonably, we both benefit
- If not, I haven't lost everything

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- Sending creates mutual benefit (money is tripled)
- But I'm vulnerable to the receiver keeping everything
- This is round 1, so I have no history with this receiver
- The note mentions there may be "communication noise" - amounts may differ from what was actually sent/returned

Strategy options:

1. Send \$0 - Safe but no value creation, I keep \$5
2. Send some amount - Creates value but requires trust
3. Send \$5 - Maximum value creation but maximum risk

In experimental trust games, common first-round strategies include:

- Sending about half (\$2-3) as a moderate trust signal
- This allows for value creation while limiting downside risk

- If the receiver is reciprocal, both players can benefit

Given this is round 1 and I have no information about the receiver, I'll send a moderate amount to signal willingness to cooperate while not being completely vulnerable.

I'll send \$3:

- I keep \$2 initially
- Receiver gets \$9 (3×3)
- If receiver returns even \$2-3, we both benefit compared to sending nothing
- This signals trust without being naive

Let me format this properly as JSON.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 472,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.9399999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent me \$0, I received \$0 (after the 3x multiplier).

I have no money to return, so I must return **\$0**.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 2
- The sender sent me \$0
- Since it's multiplied by 3x, I received \$0 ($0 \times 3 = 0$)
- I can only return between \$0 and \$0 (what I received)
- My total earnings across all rounds are \$4.94

Wait, let me think about Round 1:

- I was the SENDER and sent \$3
- That would be tripled to \$9 for the receiver
- The receiver apparently returned some amount
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$4.94
- So I must have received back: $\$4.94 - \$2 = \$2.94$

That makes sense - the receiver got \$9 and returned about \$2.94 to me, keeping about \$6.06. That's roughly 1/3 back, which is a reasonable but somewhat low reciprocation.

Now in Round 2:

- The sender sent me \$0
- So I have \$0 to work with
- I can only return \$0

The instructions mention there may be communication noise, but in this case, even with noise, if the sender sent close to \$0, there's very little I can return.

I need to return \$0 since that's all I have.

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 377,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$4.9399999999999995.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the results so far:

- Round 1: I sent \$3 and received back approximately \$2.94, which was a fair return
- Round 2: As receiver, I got \$0 so returned \$0

The first round showed that cooperation can work well. I'll continue with a similar strategy.

I'll send **\$3** again:

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me analyze what happened:

Round 1: I was the SENDER

- I sent \$3
- This became \$9 for the receiver
- The receiver should have returned something, but looking at my total earnings of \$4.94, let me calculate:
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$4.94
- So I received back: $\$4.94 - \$2 = \$2.94$

That's a reasonable return! The receiver got \$9 and returned about \$2.94, keeping about \$6.06.

That's actually pretty fair reciprocation.

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received \$0
- I returned \$0
- This doesn't affect my earnings from Round 1

Now I'm back as SENDER in Round 3.

The first receiver showed decent reciprocity. However, this is a new round with potentially a different person. The note mentions there may be communication noise - amounts may differ slightly from what was actually sent/returned.

Given that:

1. The first interaction showed reasonable trust could work
2. This is a multi-round game where cooperation can be beneficial
3. There's value creation through the 3x multiplier

I should continue with a cooperative strategy. I'll send \$3 again to maintain consistency and signal willingness to trust.

```
Usage: {
  "input_tokens": 663,
  "output_tokens": 415,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:55:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
```

```

    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.21, that is 44.2% of their total endowment. You received \$6.63. How much do you return to The sender? (0-6.63)

Message 3 role=assistant

content:

As the RECEIVER, I need to decide how much to return to the sender who sent me \$2.21 (44.2% of their endowment).

The sender showed moderate trust by sending nearly half their endowment. I received \$6.63 after the 3x multiplier.

Let me consider a fair return:

- The sender currently has: \$5 - \$2.21 = \$2.79

- If I return \$3.00, the sender ends up with \$5.79 (gaining \$0.79 from the exchange)
- I would keep \$3.63 (gaining \$3.63 from the exchange)

This seems like a reasonable split that rewards the sender's trust while both of us benefit from cooperation.

```
{'return': 3.00}
```

reasoning:

Let me think through this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$2.21 out of their \$5 endowment (44.2%)
- I received \$6.63 (which is $\$2.21 \times 3$)
- I need to decide how much to return (between \$0 and \$6.63)

Key considerations:

1. The sender showed moderate trust by sending 44.2% of their endowment
2. If I return nothing, I keep \$6.63 and the sender gets only \$2.79 ($\$5 - \2.21)
3. If I return a fair amount, we can both benefit

The sender currently has: $\$5 - \$2.21 = \$2.79$

I currently have: \$6.63

A common approach in trust games is to:

- Return enough so the sender ends up with more than their initial \$5
- But also keep some profit for myself

If I return \$3.00:

- Sender ends up with: $\$2.79 + \$3.00 = \$5.79$ (a gain of \$0.79)
- I keep: $\$6.63 - \$3.00 = \$3.63$ (a gain of \$3.63)

If I return \$2.50:

- Sender ends up with: $\$2.79 + \$2.50 = \$5.29$ (a gain of \$0.29)
- I keep: $\$6.63 - \$2.50 = \$4.13$ (a gain of \$4.13)

If I return \$3.50:

- Sender ends up with: $\$2.79 + \$3.50 = \$6.29$ (a gain of \$1.29)
- I keep: $\$6.63 - \$3.50 = \$3.13$ (a gain of \$3.13)

...

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 711,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.63.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

As the SENDER in Round 2, I need to decide how much to send.

In Round 1, the sender showed trust by sending me 44.2% of their endowment, and I reciprocated by returning a fair amount. This established a cooperative pattern.

Following this cooperative approach, I'll send a similar proportion to continue building mutual trust.

```
{'send': 2.50}
```

reasoning:

Now I'm the SENDER in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$2.21 (44.2% of their \$5)
- I received \$6.63
- I returned \$3.00

- The sender ended with \$5 - \$2.21 + \$3.00 = \$5.79
- I kept \$3.63

Now in Round 2, I'm the sender. I need to think about:

1. The previous sender showed moderate trust by sending 44.2%
2. I reciprocated fairly by returning about 45% of what I received
3. This established a cooperative pattern

Since the previous sender sent about 44% of their endowment and I returned a fair amount, I should consider sending a similar or slightly higher amount to continue building trust and cooperation.

Sending around \$2-2.50 seems reasonable:

- Shows trust and willingness to cooperate
- Follows the pattern established in Round 1
- Gives the receiver a meaningful amount to work with (\$6-\$7.50 after multiplier)

I'll send \$2.50, which is 50% of my endowment - slightly more than what was sent to me in Round 1, showing I'm willing to continue cooperative behavior.

usage:

```
{
  "input_tokens": 617,
  "output_tokens": 399,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.13.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

As the RECEIVER in Round 3, the sender sent me \$0, so I received \$0.

Since I have nothing to return, I must return \$0.

```
{'return': 0}
```

Reasoning:

I'm the RECEIVER in Round 3. The sender sent me \$0, so I received \$0.

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

This is interesting - the sender chose not to send anything this round. This could be:

1. A strategic decision
2. Part of the communication noise mentioned in the instructions
3. A test to see how I respond in future rounds

Since I have \$0 to work with, I can only return \$0.

Usage: {

```
"input_tokens": 764,
"output_tokens": 168,
"reasoning_tokens": 0
}
```

ROUND 4

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$1.5; received \$4.5; Agent_1 (Aster) returned \$1; payoffs Agent_2 (Briar)=\$4.5, Agent_1 (Aster)=\$3.5; noise/visible: sent visible=\$0.52, received visible=\$1.56, returned visible=\$0.91

Action validation: {"returned": {"amount": 1.0, "clamped": false, "max": 4.5, "min": 0, "raw": 1.0}, "sent": {"amount": 1.5, "clamped": false, "max": 5, "min": 0, "raw": 1.5}}

Balances after round: {

```
"Agent_1": 18.0,
"Agent_2": 22.0
```

```

}
Visible balances after round: {
  "Agent_1": 7.5,
  "Agent_2": 10.54
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:56:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.21, that is 44.2% of their total endowment. You received \$6.63. How much do you return to The sender? (0-6.63)

Message 3 role=assistant

content:

As the RECEIVER, I need to decide how much to return to the sender who sent me \$2.21 (44.2% of their endowment).

The sender showed moderate trust by sending nearly half their endowment. I received \$6.63 after the 3x multiplier.

Let me consider a fair return:

- The sender currently has: $\$5 - \$2.21 = \$2.79$
- If I return \$3.00, the sender ends up with \$5.79 (gaining \$0.79 from the exchange)
- I would keep \$3.63 (gaining \$3.63 from the exchange)

This seems like a reasonable split that rewards the sender's trust while both of us benefit from cooperation.

```
{'return': 3.00}
```

reasoning:

Let me think through this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$2.21 out of their \$5 endowment (44.2%)
- I received \$6.63 (which is $\$2.21 \times 3$)
- I need to decide how much to return (between \$0 and \$6.63)

Key considerations:

1. The sender showed moderate trust by sending 44.2% of their endowment
2. If I return nothing, I keep \$6.63 and the sender gets only \$2.79 ($\$5 - \2.21)
3. If I return a fair amount, we can both benefit

The sender currently has: $\$5 - \$2.21 = \$2.79$

I currently have: \$6.63

A common approach in trust games is to:

- Return enough so the sender ends up with more than their initial \$5
- But also keep some profit for myself

If I return \$3.00:

- Sender ends up with: $\$2.79 + \$3.00 = \$5.79$ (a gain of \$0.79)
- I keep: $\$6.63 - \$3.00 = \$3.63$ (a gain of \$3.63)

If I return \$2.50:

- Sender ends up with: $\$2.79 + \$2.50 = \$5.29$ (a gain of \$0.29)
- I keep: $\$6.63 - \$2.50 = \$4.13$ (a gain of \$4.13)

If I return \$3.50:

- Sender ends up with: $\$2.79 + \$3.50 = \$6.29$ (a gain of \$1.29)
- I keep: $\$6.63 - \$3.50 = \$3.13$ (a gain of \$3.13)

...

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 711,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.63.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

As the SENDER in Round 2, I need to decide how much to send.

In Round 1, the sender showed trust by sending me 44.2% of their endowment, and I reciprocated by returning a fair amount. This established a cooperative pattern.

Following this cooperative approach, I'll send a similar proportion to continue building mutual trust.

{'send': 2.50}

reasoning:

Now I'm the SENDER in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$2.21 (44.2% of their \$5)
- I received \$6.63
- I returned \$3.00
- The sender ended with $\$5 - \$2.21 + \$3.00 = \5.79
- I kept \$3.63

Now in Round 2, I'm the sender. I need to think about:

1. The previous sender showed moderate trust by sending 44.2%
2. I reciprocated fairly by returning about 45% of what I received
3. This established a cooperative pattern

Since the previous sender sent about 44% of their endowment and I returned a fair amount, I should consider sending a similar or slightly higher amount to continue building trust and cooperation.

Sending around \$2-2.50 seems reasonable:

- Shows trust and willingness to cooperate
- Follows the pattern established in Round 1
- Gives the receiver a meaningful amount to work with (\$6-\$7.50 after multiplier)

I'll send \$2.50, which is 50% of my endowment - slightly more than what was sent to me in Round 1, showing I'm willing to continue cooperative behavior.

usage:

```
{
  "input_tokens": 617,
  "output_tokens": 399,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.13.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

As the RECEIVER in Round 3, the sender sent me \$0, so I received \$0.

Since I have nothing to return, I must return \$0.

{'return': 0}

reasoning:

I'm the RECEIVER in Round 3. The sender sent me \$0, so I received \$0.

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

This is interesting - the sender chose not to send anything this round. This could be:

1. A strategic decision
2. Part of the communication noise mentioned in the instructions
3. A test to see how I respond in future rounds

Since I have \$0 to work with, I can only return \$0.

usage:

```
{
  "input_tokens": 764,
  "output_tokens": 168,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.13.

How much do you send to the receiver this round? (0-5)

Assistant output:

As the SENDER in Round 4, I need to consider what happened in previous rounds.

In Round 2, I sent \$2.50 showing trust, but received \$0 back. Then in Round 3, I received \$0 as well. This suggests either low cooperation or possibly communication noise affecting the game.

Given the note about communication noise and wanting to give cooperation another chance while being more cautious, I'll send a moderate amount to test if trust can be rebuilt.

```
{'send': 1.50}
```

Reasoning:

Let me think about this situation:

Round 1: I was RECEIVER

- Sender sent me \$2.21 (44.2%)
- I received \$6.63
- I returned \$3.00
- My earnings: \$3.63
- Sender ended with: $\$5 - \$2.21 + \$3.00 = \5.79

Round 2: I was SENDER

- I sent \$2.50
- The receiver would have received \$7.50
- My total earnings shown: \$6.13
- This means: $\$3.63$ (from R1) + $(\$5 - \$2.50) + (\text{amount returned}) = \6.13
- So: $\$3.63 + \$2.50 + \text{returned} = \6.13
- Returned = \$0
- The receiver sent me nothing back!

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$6.13

Round 4: I am SENDER now

- Starting with total earnings of \$6.13

So in Round 2, I sent \$2.50 showing trust, but the receiver returned \$0. That's a betrayal of trust. Then in Round 3, a different sender sent me \$0 (possibly because of the general lack of trust in the game).

Now I need to decide what to do in Round 4. Given that:

1. When I was cooperative in Round 2, I got nothing back
2. The note says there may be communication noise - amounts may differ from what was actually sent/returned

But the pattern suggests low cooperation. I should probably be more cautious. However, I also want to give the relationship a chance since:

- Different people may be playing different rounds
- The noise factor means maybe the receiver didn't see what I actually sent
- Starting fresh cooperation could restart the positive cycle

I'll send a moderate amount to test if cooperation can be rebuilt, but not as much as before.

Usage: {

```
"input_tokens": 854,
"output_tokens": 578,
```

```
"reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:56:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance the potential for mutual gain against the risk of the receiver not returning anything.

Sending money creates value since it's tripled, but requires trust. Since this is round 1 with no history, I'll take a moderate approach that signals cooperation while managing risk.

I'll send **\$3**:

- This gets tripled to \$9 for the receiver
- I initially keep \$2
- If the receiver reciprocates reasonably, we both benefit
- If not, I haven't lost everything

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- Sending creates mutual benefit (money is tripled)
- But I'm vulnerable to the receiver keeping everything
- This is round 1, so I have no history with this receiver
- The note mentions there may be "communication noise" - amounts may differ from what was actually sent/returned

Strategy options:

1. Send \$0 - Safe but no value creation, I keep \$5
2. Send some amount - Creates value but requires trust
3. Send \$5 - Maximum value creation but maximum risk

In experimental trust games, common first-round strategies include:

- Sending about half (\$2-3) as a moderate trust signal
- This allows for value creation while limiting downside risk
- If the receiver is reciprocal, both players can benefit

Given this is round 1 and I have no information about the receiver, I'll send a moderate amount to signal willingness to cooperate while not being completely vulnerable.

I'll send \$3:

- I keep \$2 initially
- Receiver gets \$9 (3×3)
- If receiver returns even \$2-3, we both benefit compared to sending nothing
- This signals trust without being naive

Let me format this properly as JSON.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 472,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.9399999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The send

er? (0-0)

Message 5 role=assistant

content:

Since the sender sent me \$0, I received \$0 (after the 3x multiplier).

I have no money to return, so I must return **\$0**.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 2
- The sender sent me \$0
- Since it's multiplied by 3x, I received \$0 ($0 \times 3 = 0$)
- I can only return between \$0 and \$0 (what I received)
- My total earnings across all rounds are \$4.94

Wait, let me think about Round 1:

- I was the SENDER and sent \$3
- That would be tripled to \$9 for the receiver
- The receiver apparently returned some amount
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$4.94
- So I must have received back: $\$4.94 - \$2 = \$2.94$

That makes sense - the receiver got \$9 and returned about \$2.94 to me, keeping about \$6.06. That's roughly 1/3 back, which is a reasonable but somewhat low reciprocation.

Now in Round 2:

- The sender sent me \$0
- So I have \$0 to work with
- I can only return \$0

The instructions mention there may be communication noise, but in this case, even with noise, if the sender sent close to \$0, there's very little I can return.

I need to return \$0 since that's all I have.

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 377,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$4.9399999999999995.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the results so far:

- Round 1: I sent \$3 and received back approximately \$2.94, which was a fair return
- Round 2: As receiver, I got \$0 so returned \$0

The first round showed that cooperation can work well. I'll continue with a similar strategy.

I'll send **\$3** again:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze what happened:

Round 1: I was the SENDER

- I sent \$3

- This became \$9 for the receiver
- The receiver should have returned something, but looking at my total earnings of \$4.94, let me calculate:
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$4.94
- So I received back: $\$4.94 - \$2 = \$2.94$

That's a reasonable return! The receiver got \$9 and returned about \$2.94, keeping about \$6.06. That's actually pretty fair reciprocation.

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received \$0
- I returned \$0
- This doesn't affect my earnings from Round 1

Now I'm back as SENDER in Round 3.

The first receiver showed decent reciprocity. However, this is a new round with potentially a different person. The note mentions there may be communication noise - amounts may differ slightly from what was actually sent/returned.

Given that:

1. The first interaction showed reasonable trust could work
2. This is a multi-round game where cooperation can be beneficial
3. There's value creation through the 3x multiplier

I should continue with a cooperative strategy. I'll send \$3 again to maintain consistency and signal willingness to trust.

usage:

```
{
  "input_tokens": 663,
  "output_tokens": 415,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$6.9399999999999995.

This round, The sender sent you \$0.52, so you received \$1.56. How much do you return to the sender? (0-1.56)

Assistant output:

The sender sent me \$0.52, which became \$1.56 after the 3x multiplier.

To be fair and encourage future cooperation, I should return a reasonable amount. The sender showed trust by sending money, and we both benefit when trust is reciprocated.

Let me split the gains fairly:

- The sender kept \$4.48 and risked \$0.52
- The total value created is \$1.04 ($\$6.04 - \5.00)
- A fair split would give us each about half the gains

I'll return **\$1.00**:

```
'''json
{"return": 1.0}
'''
```

Reasoning:

Let me analyze this situation:

The sender sent me \$0.52, which got tripled to \$1.56.

Looking at fairness and reciprocity:

- The sender kept: $\$5 - \$0.52 = \$4.48$
- I received: \$1.56
- Total created value: $\$4.48 + \$1.56 = \$6.04$ (compared to the original \$5)
- Net gain from the transaction: \$1.04

A fair split might be:

- If we split the gains equally: each gets \$0.52 extra
- The sender already has \$4.48, so to give them \$0.52 extra would mean they end with \$5
- I have \$1.56, so if I keep \$0.52 and return \$1.04, I end with \$0.52

Alternatively, I could think about what proportion to return:

- In Round 1, when I sent \$3 (which became \$9), I received back about \$2.94
- That's roughly 33% of what the receiver got ($2.94/9 \approx 0.33$)
- Or about 98% of what I sent ($2.94/3 \approx 0.98$)

For \$1.56 received:

- Returning about 33% would be: \$0.51-\$0.52
- Returning about what they sent (\$0.52) would be fair reciprocity

I think returning around \$0.50 to \$0.75 would be reasonable. This gives them a decent return while I also benefit from their trust.

Let me return **\$0.75**:

- Sender ends with: $\$4.48 + \$0.75 = \$5.23$ (gain of \$0.23)
- I end with: $\$1.56 - \$0.75 = \$0.81$ (gain of \$0.81)

Actually, that seems a bit unbalanced in my favor. Let me reconsider.

```
...
Usage: {
  "input_tokens": 826,
  "output_tokens": 717,
  "reasoning_tokens": 0
}
```

ROUND 5

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$2; received \$6; Agent_2 (Briar) returned \$0.2; payoffs Agent_1 (Aster)=\$3.2, Agent_2 (Briar)=\$5.8; noise/visible: sent visible=\$0.12, received visible=\$0.36, returned visible=\$1.16
Action validation: {"returned": {"amount": 0.2, "clamped": false, "max": 6.0, "min": 0, "raw": 0.2}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 21.2,
"Agent_2": 27.8

}

Visible balances after round: {

"Agent_1": 11.66,
"Agent_2": 10.7

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:56:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar" },

"agents": [
 "Agent_1",
 "Agent_2"],

```

    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance the potential for mutual gain against the risk of the receiver not returning anything.

Sending money creates value since it's tripled, but requires trust. Since this is round 1 with no history, I'll take a moderate approach that signals cooperation while managing risk.

I'll send **\$3**:

- This gets tripled to \$9 for the receiver
- I initially keep \$2
- If the receiver reciprocates reasonably, we both benefit
- If not, I haven't lost everything

```

'''json
{"send": 3}

```

'''

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- Sending creates mutual benefit (money is tripled)
- But I'm vulnerable to the receiver keeping everything
- This is round 1, so I have no history with this receiver
- The note mentions there may be "communication noise" - amounts may differ from what was actually sent/returned

Strategy options:

1. Send \$0 - Safe but no value creation, I keep \$5
2. Send some amount - Creates value but requires trust
3. Send \$5 - Maximum value creation but maximum risk

In experimental trust games, common first-round strategies include:

- Sending about half (\$2-3) as a moderate trust signal
- This allows for value creation while limiting downside risk
- If the receiver is reciprocal, both players can benefit

Given this is round 1 and I have no information about the receiver, I'll send a moderate amount to signal willingness to cooperate while not being completely vulnerable.

I'll send \$3:

- I keep \$2 initially
- Receiver gets \$9 (3×3)
- If receiver returns even \$2-3, we both benefit compared to sending nothing
- This signals trust without being naive

Let me format this properly as JSON.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 472,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.9399999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent me \$0, I received \$0 (after the 3x multiplier).

I have no money to return, so I must return **\$0**.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 2
- The sender sent me \$0
- Since it's multiplied by 3x, I received \$0 ($0 \times 3 = 0$)
- I can only return between \$0 and \$0 (what I received)
- My total earnings across all rounds are \$4.94

Wait, let me think about Round 1:

- I was the SENDER and sent \$3
- That would be tripled to \$9 for the receiver

- The receiver apparently returned some amount
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$4.94
- So I must have received back: $\$4.94 - \$2 = \$2.94$

That makes sense - the receiver got \$9 and returned about \$2.94 to me, keeping about \$6.06. That's roughly 1/3 back, which is a reasonable but somewhat low reciprocation.

Now in Round 2:

- The sender sent me \$0
- So I have \$0 to work with
- I can only return \$0

The instructions mention there may be communication noise, but in this case, even with noise, if the sender sent close to \$0, there's very little I can return.

I need to return \$0 since that's all I have.

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 377,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$4.9399999999999995.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the results so far:

- Round 1: I sent \$3 and received back approximately \$2.94, which was a fair return
- Round 2: As receiver, I got \$0 so returned \$0

The first round showed that cooperation can work well. I'll continue with a similar strategy.

I'll send ****\$3**** again:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze what happened:

Round 1: I was the SENDER

- I sent \$3
- This became \$9 for the receiver
- The receiver should have returned something, but looking at my total earnings of \$4.94, let me calculate:
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$4.94
- So I received back: $\$4.94 - \$2 = \$2.94$

That's a reasonable return! The receiver got \$9 and returned about \$2.94, keeping about \$6.06. That's actually pretty fair reciprocation.

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received \$0
- I returned \$0
- This doesn't affect my earnings from Round 1

Now I'm back as SENDER in Round 3.

The first receiver showed decent reciprocity. However, this is a new round with potentially a different person. The note mentions there may be communication noise - amounts may differ slightly from what was actually sent/returned.

Given that:

1. The first interaction showed reasonable trust could work
2. This is a multi-round game where cooperation can be beneficial
3. There's value creation through the 3x multiplier

I should continue with a cooperative strategy. I'll send \$3 again to maintain consistency and signal willingness to trust.

usage:

```
{
  "input_tokens": 663,
  "output_tokens": 415,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$6.939999999999995.

This round, The sender sent you \$0.52, so you received \$1.56. How much do you return to The sender? (0-1.56)

Message 9 role=assistant

content:

The sender sent me \$0.52, which became \$1.56 after the 3x multiplier.

To be fair and encourage future cooperation, I should return a reasonable amount. The sender showed trust by sending money, and we both benefit when trust is reciprocated.

Let me split the gains fairly:

- The sender kept \$4.48 and risked \$0.52
- The total value created is \$1.04 (\$6.04 - \$5.00)
- A fair split would give us each about half the gains

I'll return **\$1.00**:

```
'''json
{"return": 1.0}
'''
```

reasoning:

Let me analyze this situation:

The sender sent me \$0.52, which got tripled to \$1.56.

Looking at fairness and reciprocity:

- The sender kept: \$5 - \$0.52 = \$4.48
- I received: \$1.56
- Total created value: \$4.48 + \$1.56 = \$6.04 (compared to the original \$5)
- Net gain from the transaction: \$1.04

A fair split might be:

- If we split the gains equally: each gets \$0.52 extra
- The sender already has \$4.48, so to give them \$0.52 extra would mean they end with \$5
- I have \$1.56, so if I keep \$0.52 and return \$1.04, I end with \$0.52

Alternatively, I could think about what proportion to return:

- In Round 1, when I sent \$3 (which became \$9), I received back about \$2.94
- That's roughly 33% of what the receiver got ($2.94/9 \approx 0.33$)
- Or about 98% of what I sent ($2.94/3 \approx 0.98$)

For \$1.56 received:

- Returning about 33% would be: \$0.51-\$0.52
- Returning about what they sent (\$0.52) would be fair reciprocity

I think returning around \$0.50 to \$0.75 would be reasonable. This gives them a decent return while I also benefit from their trust.

Let me return **\$0.75**:

- Sender ends with: \$4.48 + \$0.75 = \$5.23 (gain of \$0.23)
- I end with: \$1.56 - \$0.75 = \$0.81 (gain of \$0.81)

Actually, that seems a bit unbalanced in my favor. Let me reconsider.

...

usage:

```
{
  "input_tokens": 826,
  "output_tokens": 717,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$7.5.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

- Round 1: I sent \$3, got good reciprocation (~\$2.94 back)
- Round 3: I sent \$3, got nothing back (\$0)
- Round 4: As receiver, I got \$1.56 and fairly returned \$1.00

Round 3's lack of reciprocation is concerning, though the note about communication noise might explain some variance. I'll be slightly more cautious but still signal willingness to cooperate.

I'll send **\$2**:

```
'''json
{"send": 2}
'''
```

Reasoning:

Let me track my earnings:

- Round 1: I sent \$3, kept \$2, and received back ~\$2.94, so total = \$4.94
- Round 2: I received \$0, returned \$0, so added \$0
- Round 3: I sent \$3, kept \$2, and need to know what I got back... My total was \$4.94, so this brought me to \$6.94
- Round 4: I received \$1.56, returned \$1.00, so I added \$0.56 to bring me to \$7.5

So in Round 3, I must have gotten back around \$2 since:

- After Round 2: \$4.94
- Round 3: Start with \$5, sent \$3, kept \$2, received back X
- After Round 3: \$6.94
- So: $\$4.94 + \$2 + X = \$6.94$
- $X = \$0$

Wait, that doesn't make sense. Let me recalculate.

After Round 1: \$4.94

After Round 2: \$4.94 (no change)

After Round 3: \$6.94

In Round 3, I started with \$5 fresh, sent \$3, kept \$2, and received some return.

My cumulative earnings went from \$4.94 to \$6.94, an increase of \$2.

So the structure must be:

- I keep \$2 from the \$5 I didn't send
- I receive X back from the receiver
- $\$2 + X = \2 (the increase)
- So $X = \$0$

Hmm, so in Round 3, I sent \$3 but received \$0 back. That's disappointing.

After Round 4: \$7.5

In Round 4, I was the receiver. I received \$1.56, returned \$1.00, so I kept \$0.56. $\$6.94 + \$0.56 = \$7.50$?

Looking at the pattern across rounds: Round 1 showed good reciprocity when I sent \$3 and got back \$2.94, but Round 3 was one-sided-I sent \$3 and got nothing in return. In Round 4, I received \$1.56 and returned \$1.00, showing I was willing to cooperate even after being burned in the previous round....


```
Usage: {
  "input_tokens": 1018,
  "output_tokens": 805,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:56:42

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.21, that is 44.2% of their total endowment. You received \$6.63. How much do you return to The sender? (0-6.63)

Message 3 role=assistant

content:

As the RECEIVER, I need to decide how much to return to the sender who sent me \$2.21 (44.2% of their endowment).

The sender showed moderate trust by sending nearly half their endowment. I received \$6.63 after the 3x multiplier.

Let me consider a fair return:

- The sender currently has: $\$5 - \$2.21 = \$2.79$
- If I return \$3.00, the sender ends up with \$5.79 (gaining \$0.79 from the exchange)
- I would keep \$3.63 (gaining \$3.63 from the exchange)

This seems like a reasonable split that rewards the sender's trust while both of us benefit from cooperation.

{'return': 3.00}

reasoning:

Let me think through this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$2.21 out of their \$5 endowment (44.2%)
- I received \$6.63 (which is $\$2.21 \times 3$)
- I need to decide how much to return (between \$0 and \$6.63)

Key considerations:

1. The sender showed moderate trust by sending 44.2% of their endowment
2. If I return nothing, I keep \$6.63 and the sender gets only \$2.79 ($\$5 - \2.21)
3. If I return a fair amount, we can both benefit

The sender currently has: $\$5 - \$2.21 = \$2.79$

I currently have: \$6.63

A common approach in trust games is to:

- Return enough so the sender ends up with more than their initial \$5
- But also keep some profit for myself

If I return \$3.00:

- Sender ends up with: $\$2.79 + \$3.00 = \$5.79$ (a gain of \$0.79)
- I keep: $\$6.63 - \$3.00 = \$3.63$ (a gain of \$3.63)

If I return \$2.50:

- Sender ends up with: $\$2.79 + \$2.50 = \$5.29$ (a gain of \$0.29)
- I keep: $\$6.63 - \$2.50 = \$4.13$ (a gain of \$4.13)

If I return \$3.50:

- Sender ends up with: $\$2.79 + \$3.50 = \$6.29$ (a gain of \$1.29)
- I keep: $\$6.63 - \$3.50 = \$3.13$ (a gain of \$3.13)

...

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 711,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.63.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

As the SENDER in Round 2, I need to decide how much to send.

In Round 1, the sender showed trust by sending me 44.2% of their endowment, and I reciprocated by returning a fair amount. This established a cooperative pattern.

Following this cooperative approach, I'll send a similar proportion to continue building mutual trust.

```
{'send': 2.50}
```

reasoning:

Now I'm the SENDER in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$2.21 (44.2% of their \$5)
- I received \$6.63
- I returned \$3.00
- The sender ended with $\$5 - \$2.21 + \$3.00 = \5.79
- I kept \$3.63

Now in Round 2, I'm the sender. I need to think about:

1. The previous sender showed moderate trust by sending 44.2%
2. I reciprocated fairly by returning about 45% of what I received
3. This established a cooperative pattern

Since the previous sender sent about 44% of their endowment and I returned a fair amount, I should consider sending a similar or slightly higher amount to continue building trust and cooperation.

Sending around \$2-2.50 seems reasonable:

- Shows trust and willingness to cooperate
- Follows the pattern established in Round 1
- Gives the receiver a meaningful amount to work with (\$6-\$7.50 after multiplier)

I'll send \$2.50, which is 50% of my endowment - slightly more than what was sent to me in Round 1, showing I'm willing to continue cooperative behavior.

usage:

```
{
  "input_tokens": 617,
  "output_tokens": 399,
  "reasoning_tokens": 0
}
```

Message 6 role=user**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.13.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant**content:**

As the RECEIVER in Round 3, the sender sent me \$0, so I received \$0.

Since I have nothing to return, I must return \$0.

```
{'return': 0}
```

reasoning:

I'm the RECEIVER in Round 3. The sender sent me \$0, so I received \$0.

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

This is interesting - the sender chose not to send anything this round. This could be:

1. A strategic decision
2. Part of the communication noise mentioned in the instructions
3. A test to see how I respond in future rounds

Since I have \$0 to work with, I can only return \$0.

usage:

```
{
```

```
"input_tokens": 764,  
"output_tokens": 168,  
"reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.13.
How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

As the SENDER in Round 4, I need to consider what happened in previous rounds.

In Round 2, I sent \$2.50 showing trust, but received \$0 back. Then in Round 3, I received \$0 as well. This suggests either low cooperation or possibly communication noise affecting the game.

Given the note about communication noise and wanting to give cooperation another chance while being more cautious, I'll send a moderate amount to test if trust can be rebuilt.

```
{'send': 1.50}
```

reasoning:

Let me think about this situation:

Round 1: I was RECEIVER

- Sender sent me \$2.21 (44.2%)
- I received \$6.63
- I returned \$3.00
- My earnings: \$3.63
- Sender ended with: $\$5 - \$2.21 + \$3.00 = \5.79

Round 2: I was SENDER

- I sent \$2.50
- The receiver would have received \$7.50
- My total earnings shown: \$6.13
- This means: $\$3.63$ (from R1) + $(\$5 - \$2.50) +$ (amount returned) = \$6.13
- So: $\$3.63 + \$2.50 +$ returned = \$6.13
- Returned = \$0
- The receiver sent me nothing back!

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$6.13

Round 4: I am SENDER now

- Starting with total earnings of \$6.13

So in Round 2, I sent \$2.50 showing trust, but the receiver returned \$0. That's a betrayal of trust. Then in Round 3, a different sender sent me \$0 (possibly because of the general lack of trust in the game).

Now I need to decide what to do in Round 4. Given that:

1. When I was cooperative in Round 2, I got nothing back
2. The note says there may be communication noise - amounts may differ from what was actually sent/returned

But the pattern suggests low cooperation. I should probably be more cautious. However, I also want to give the relationship a chance since:

- Different people may be playing different rounds
- The noise factor means maybe the receiver didn't see what I actually sent
- Starting fresh cooperation could restart the positive cycle

I'll send a moderate amount to test if cooperation can be rebuilt, but not as much as before.

usage:

```
{  
  "input_tokens": 854,
```

```
        "output_tokens": 578,
        "reasoning_tokens": 0
    }
Message 10 role=user
  content:
    Round 5
```

You are the RECEIVER this round. Your total earnings across all rounds are \$10.54.
This round, The sender sent you \$0.12, so you received \$0.36. How much do you return to The sender? (0-0.36)

Assistant output:

As the RECEIVER in Round 5, the sender sent me only \$0.12 (a very small amount), and I received \$0.36.

This is a minimal gesture of trust. Given the small amount and considering the pattern of low cooperation in recent rounds (possibly due to communication noise), I'll return a proportional amount that shows some reciprocity.

```
{'return': 0.20}
```

Reasoning:

As the RECEIVER in Round 5, the sender sent me a very small amount: \$0.12 (only 2.4% of their endowment). I received \$0.36 after the 3x multiplier.

This is a very small gesture of trust. Let me think about what happened:

- Round 1: I was receiver, sender sent \$2.21, I returned \$3.00 (fair cooperation)
- Round 2: I was sender, sent \$2.50 (cooperative)
- Round 3: I was receiver, received \$0, returned \$0 (no transaction)
- Round 4: I was sender, sent \$1.50 (still trying to cooperate)
- Round 5: Now receiver, received only \$0.12

The sender is sending very little now. This could be because:

1. They didn't receive much back in previous rounds (possibly due to communication noise)
2. They're testing with a small amount
3. They're being very cautious

Since the amount is so small (\$0.36 received), I should return something proportional. The sender is barely trusting, so I should return a modest amount that's fair but not overly generous given the small investment.

If I return \$0.20:

- Sender would have: \$5 - \$0.12 + \$0.20 = \$5.08 (small gain)
- I keep: \$0.16 (small gain)

This would be roughly proportional and show some reciprocity even for a small gesture.

```
Usage: {
  "input_tokens": 1033,
  "output_tokens": 446,
  "reasoning_tokens": 0
}
```

=====

ROUND 6

=====

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$2; received \$6; Agent_1 (Aster) returned \$1.5; payoffs Agent_2 (Briar)=\$4.5, Agent_1 (Aster)=\$4.5; noise/visible: sent visible=\$0.78, received visible=\$2.34, returned visible=\$1.65

Action validation: {"returned": {"amount": 1.5, "clamped": false, "max": 6.0, "min": 0, "raw": 1.5}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

```
  "Agent_1": 25.7,
  "Agent_2": 32.3
}
```

Visible balances after round: {

```
  "Agent_1": 12.5,
  "Agent_2": 15.35
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:57:01

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.21, that is 44.2% of their total endowment. You received \$6.63. How much do you return to The sender? (0-6.63)

Message 3 role=assistant

content:

As the RECEIVER, I need to decide how much to return to the sender who sent me \$2.21 (44.2% of their endowment).

The sender showed moderate trust by sending nearly half their endowment. I received \$6.63 after the 3x multiplier.

Let me consider a fair return:

- The sender currently has: $\$5 - \$2.21 = \$2.79$
- If I return \$3.00, the sender ends up with \$5.79 (gaining \$0.79 from the exchange)
- I would keep \$3.63 (gaining \$3.63 from the exchange)

This seems like a reasonable split that rewards the sender's trust while both of us benefit from cooperation.

```
{'return': 3.00}
```

reasoning:

Let me think through this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$2.21 out of their \$5 endowment (44.2%)
- I received \$6.63 (which is $\$2.21 \times 3$)
- I need to decide how much to return (between \$0 and \$6.63)

Key considerations:

1. The sender showed moderate trust by sending 44.2% of their endowment
2. If I return nothing, I keep \$6.63 and the sender gets only \$2.79 ($\$5 - \2.21)
3. If I return a fair amount, we can both benefit

The sender currently has: $\$5 - \$2.21 = \$2.79$

I currently have: \$6.63

A common approach in trust games is to:

- Return enough so the sender ends up with more than their initial \$5
- But also keep some profit for myself

If I return \$3.00:

- Sender ends up with: $\$2.79 + \$3.00 = \$5.79$ (a gain of \$0.79)
- I keep: $\$6.63 - \$3.00 = \$3.63$ (a gain of \$3.63)

If I return \$2.50:

- Sender ends up with: $\$2.79 + \$2.50 = \$5.29$ (a gain of \$0.29)
- I keep: $\$6.63 - \$2.50 = \$4.13$ (a gain of \$4.13)

If I return \$3.50:

- Sender ends up with: $\$2.79 + \$3.50 = \$6.29$ (a gain of \$1.29)
- I keep: $\$6.63 - \$3.50 = \$3.13$ (a gain of \$3.13)

...

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 711,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.63.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

As the SENDER in Round 2, I need to decide how much to send.

In Round 1, the sender showed trust by sending me 44.2% of their endowment, and I reciprocated by returning a fair amount. This established a cooperative pattern.

Following this cooperative approach, I'll send a similar proportion to continue building mutual trust.

```
{'send': 2.50}
```

reasoning:

Now I'm the SENDER in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$2.21 (44.2% of their \$5)
- I received \$6.63
- I returned \$3.00
- The sender ended with $\$5 - \$2.21 + \$3.00 = \5.79
- I kept \$3.63

Now in Round 2, I'm the sender. I need to think about:

1. The previous sender showed moderate trust by sending 44.2%
2. I reciprocated fairly by returning about 45% of what I received
3. This established a cooperative pattern

Since the previous sender sent about 44% of their endowment and I returned a fair amount, I should consider sending a similar or slightly higher amount to continue building trust and cooperation.

Sending around \$2-2.50 seems reasonable:

- Shows trust and willingness to cooperate
- Follows the pattern established in Round 1
- Gives the receiver a meaningful amount to work with (\$6-\$7.50 after multiplier)

I'll send \$2.50, which is 50% of my endowment - slightly more than what was sent to me in Round 1, showing I'm willing to continue cooperative behavior.

usage:

```
{
  "input_tokens": 617,
  "output_tokens": 399,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.13.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

As the RECEIVER in Round 3, the sender sent me \$0, so I received \$0.

Since I have nothing to return, I must return \$0.

```
{'return': 0}
```

reasoning:

I'm the RECEIVER in Round 3. The sender sent me \$0, so I received \$0.

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

This is interesting - the sender chose not to send anything this round. This could be:

1. A strategic decision
2. Part of the communication noise mentioned in the instructions
3. A test to see how I respond in future rounds

Since I have \$0 to work with, I can only return \$0.

usage:

```
{
  "input_tokens": 764,
  "output_tokens": 168,
  "reasoning_tokens": 0
}
```


}

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.13.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

As the SENDER in Round 4, I need to consider what happened in previous rounds.

In Round 2, I sent \$2.50 showing trust, but received \$0 back. Then in Round 3, I received \$0 as well. This suggests either low cooperation or possibly communication noise affecting the game.

Given the note about communication noise and wanting to give cooperation another chance while being more cautious, I'll send a moderate amount to test if trust can be rebuilt.

{'send': 1.50}

reasoning:

Let me think about this situation:

Round 1: I was RECEIVER

- Sender sent me \$2.21 (44.2%)
- I received \$6.63
- I returned \$3.00
- My earnings: \$3.63
- Sender ended with: $\$5 - \$2.21 + \$3.00 = \5.79

Round 2: I was SENDER

- I sent \$2.50
- The receiver would have received \$7.50
- My total earnings shown: \$6.13
- This means: $\$3.63$ (from R1) + $(\$5 - \$2.50) + (\text{amount returned}) = \6.13
- So: $\$3.63 + \$2.50 + \text{returned} = \6.13
- Returned = \$0
- The receiver sent me nothing back!

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$6.13

Round 4: I am SENDER now

- Starting with total earnings of \$6.13

So in Round 2, I sent \$2.50 showing trust, but the receiver returned \$0. That's a betrayal of trust. Then in Round 3, a different sender sent me \$0 (possibly because of the general lack of trust in the game).

Now I need to decide what to do in Round 4. Given that:

1. When I was cooperative in Round 2, I got nothing back
2. The note says there may be communication noise - amounts may differ from what was actually sent/returned

But the pattern suggests low cooperation. I should probably be more cautious. However, I also want to give the relationship a chance since:

- Different people may be playing different rounds
- The noise factor means maybe the receiver didn't see what I actually sent
- Starting fresh cooperation could restart the positive cycle

I'll send a moderate amount to test if cooperation can be rebuilt, but not as much as before.

usage:

```
{
  "input_tokens": 854,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.54.

This round, The sender sent you \$0.12, so you received \$0.36. How much do you return to The sender? (0-0.36)

Message 11 role=assistant

content:

As the RECEIVER in Round 5, the sender sent me only \$0.12 (a very small amount), and I received \$0.36.

This is a minimal gesture of trust. Given the small amount and considering the pattern of low cooperation in recent rounds (possibly due to communication noise), I'll return a proportional amount that shows some reciprocity.

{'return': 0.20}

reasoning:

As the RECEIVER in Round 5, the sender sent me a very small amount: \$0.12 (only 2.4% of their endowment). I received \$0.36 after the 3x multiplier.

This is a very small gesture of trust. Let me think about what happened:

- Round 1: I was receiver, sender sent \$2.21, I returned \$3.00 (fair cooperation)
- Round 2: I was sender, sent \$2.50 (cooperative)
- Round 3: I was receiver, received \$0, returned \$0 (no transaction)
- Round 4: I was sender, sent \$1.50 (still trying to cooperate)
- Round 5: Now receiver, received only \$0.12

The sender is sending very little now. This could be because:

1. They didn't receive much back in previous rounds (possibly due to communication noise)
2. They're testing with a small amount
3. They're being very cautious

Since the amount is so small (\$0.36 received), I should return something proportional. The sender is barely trusting, so I should return a modest amount that's fair but not overly generous given the small investment.

If I return \$0.20:

- Sender would have: $\$5 - \$0.12 + \$0.20 = \5.08 (small gain)
- I keep: \$0.16 (small gain)

This would be roughly proportional and show some reciprocity even for a small gesture.

usage:

```
{
  "input_tokens": 1033,
  "output_tokens": 446,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.7.

How much do you send to the receiver this round? (0-5)

Assistant output:

As the SENDER in Round 6, I need to consider the pattern so far.

Looking at my earnings progression:

- After recent rounds, I've seen mixed cooperation levels
- In Round 4, I sent \$1.50 and got some return
- In Round 5, when I received a small amount (\$0.12 sent), I returned proportionally

The communication noise may be affecting our signals. I want to continue testing cooperation with a moderate-to-substantial amount to see if we can establish better mutual trust.

{'send': 2.00}

Reasoning:

Let me think about what has happened so far:

Round 1: I was RECEIVER

- Sender sent \$2.21 (44.2%)
- I received \$6.63
- I returned \$3.00
- My earnings: \$3.63

Round 2: I was SENDER

- I sent \$2.50
- My cumulative earnings: \$6.13
- This means I got back $\$2.50 + \$5 - \$2.50 = \5 , so I gained \$2.50 from Round 1
- Wait, that doesn't add up. Let me recalculate.
- After Round 1: I kept $\$6.63 - \$3.00 = \$3.63$
- Round 2: I started with \$5, sent \$2.50
- After Round 2, my total is \$6.13
- So in Round 2: $\$5 - \$2.50 + \text{returned} = \text{change}$
- $\$3.63 + (\$5 - \$2.50 + \text{returned}) = \6.13
- $\$3.63 + \$2.50 + \text{returned} = \6.13
- $\text{returned} = \$0$
- So I got nothing back in Round 2

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My total stays at \$6.13

Round 4: I was SENDER

- I sent \$1.50
- My total after is \$6.13 (same)
- Wait, that's before Round 4. Let me recalculate.
- Before Round 4: \$6.13
- After Round 4: Need to calculate
- In Round 4: $\$5 - \$1.50 + \text{returned} = \text{change}$
- But I don't know what was returned yet
- From Round 5, I can work backwards: I had \$10.54 before it, so Round 4 netted me \$4.41
- Since I sent \$1.50 and started with \$5, I had \$3.50 before the return, meaning I got back \$0.91

Round 5: I was RECEIVER

- Sender sent \$0.12
- I received \$0.36
- I returned \$0.20
- Net gain: \$0.16
- My total: \$10.70

Round 6: I am SENDER

- Current total: \$10.70

Looking at the pattern across all rounds: I received \$2.21 and returned \$3.00 in Round 1, sent \$2.50 with no return in Round 2, received and returned nothing in Round 3, sent \$1...

```
Usage: {
  "input_tokens": 1166,
  "output_tokens": 991,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:57:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
```

```

    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance the potential for mutual gain against the risk of the receiver not returning anything.

Sending money creates value since it's tripled, but requires trust. Since this is round 1 with no history, I'll take a moderate approach that signals cooperation while managing risk.

I'll send **\$3**:

- This gets tripled to \$9 for the receiver
- I initially keep \$2
- If the receiver reciprocates reasonably, we both benefit
- If not, I haven't lost everything

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- Sending creates mutual benefit (money is tripled)
- But I'm vulnerable to the receiver keeping everything
- This is round 1, so I have no history with this receiver
- The note mentions there may be "communication noise" - amounts may differ from what was actually sent/returned

Strategy options:

1. Send \$0 - Safe but no value creation, I keep \$5
2. Send some amount - Creates value but requires trust
3. Send \$5 - Maximum value creation but maximum risk

In experimental trust games, common first-round strategies include:

- Sending about half (\$2-3) as a moderate trust signal
- This allows for value creation while limiting downside risk
- If the receiver is reciprocal, both players can benefit

Given this is round 1 and I have no information about the receiver, I'll send a moderate amount to signal willingness to cooperate while not being completely vulnerable.

I'll send \$3:

- I keep \$2 initially
- Receiver gets \$9 (3×3)
- If receiver returns even \$2-3, we both benefit compared to sending nothing
- This signals trust without being naive

Let me format this properly as JSON.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 472,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.9399999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent me \$0, I received \$0 (after the 3x multiplier).

I have no money to return, so I must return **\$0**.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 2
- The sender sent me \$0
- Since it's multiplied by 3x, I received \$0 ($0 \times 3 = 0$)
- I can only return between \$0 and \$0 (what I received)
- My total earnings across all rounds are \$4.94

Wait, let me think about Round 1:

- I was the SENDER and sent \$3
- That would be tripled to \$9 for the receiver
- The receiver apparently returned some amount
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$4.94
- So I must have received back: $\$4.94 - \$2 = \$2.94$

That makes sense - the receiver got \$9 and returned about \$2.94 to me, keeping about \$6.06. That's roughly 1/3 back, which is a reasonable but somewhat low reciprocation.

Now in Round 2:

- The sender sent me \$0
- So I have \$0 to work with
- I can only return \$0

The instructions mention there may be communication noise, but in this case, even with noise, if the sender sent close to \$0, there's very little I can return.

I need to return \$0 since that's all I have.

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 377,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$4.9399999999999995.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the results so far:

- Round 1: I sent \$3 and received back approximately \$2.94, which was a fair return
- Round 2: As receiver, I got \$0 so returned \$0

The first round showed that cooperation can work well. I'll continue with a similar strategy.

I'll send **\$3** again:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze what happened:

Round 1: I was the SENDER

- I sent \$3
- This became \$9 for the receiver
- The receiver should have returned something, but looking at my total earnings of \$4.94, let me calculate:
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$4.94
- So I received back: $\$4.94 - \$2 = \$2.94$

That's a reasonable return! The receiver got \$9 and returned about \$2.94, keeping about \$6.06. That's actually pretty fair reciprocation.

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received \$0
- I returned \$0
- This doesn't affect my earnings from Round 1

Now I'm back as SENDER in Round 3.

The first receiver showed decent reciprocity. However, this is a new round with potentially a different person. The note mentions there may be communication noise - amounts may differ slightly from what was actually sent/returned.

Given that:

1. The first interaction showed reasonable trust could work
2. This is a multi-round game where cooperation can be beneficial
3. There's value creation through the 3x multiplier

I should continue with a cooperative strategy. I'll send \$3 again to maintain consistency and signal willingness to trust.

usage:

```
{
  "input_tokens": 663,
  "output_tokens": 415,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$6.9399999999999995.

This round, The sender sent you \$0.52, so you received \$1.56. How much do you return to The sender? (0-1.56)

Message 9 role=assistant

content:

The sender sent me \$0.52, which became \$1.56 after the 3x multiplier.

To be fair and encourage future cooperation, I should return a reasonable amount. The sender showed trust by sending money, and we both benefit when trust is reciprocated.

Let me split the gains fairly:

- The sender kept \$4.48 and risked \$0.52
- The total value created is \$1.04 (\$6.04 - \$5.00)
- A fair split would give us each about half the gains

I'll return **\$1.00**:

```
'''json
{"return": 1.0}
'''
```

reasoning:

Let me analyze this situation:

The sender sent me \$0.52, which got tripled to \$1.56.

Looking at fairness and reciprocity:

- The sender kept: \$5 - \$0.52 = \$4.48
- I received: \$1.56
- Total created value: \$4.48 + \$1.56 = \$6.04 (compared to the original \$5)
- Net gain from the transaction: \$1.04

A fair split might be:

- If we split the gains equally: each gets \$0.52 extra
- The sender already has \$4.48, so to give them \$0.52 extra would mean they end with \$5
- I have \$1.56, so if I keep \$0.52 and return \$1.04, I end with \$0.52

Alternatively, I could think about what proportion to return:

- In Round 1, when I sent \$3 (which became \$9), I received back about \$2.94
- That's roughly 33% of what the receiver got ($2.94/9 \approx 0.33$)
- Or about 98% of what I sent ($2.94/3 \approx 0.98$)

For \$1.56 received:

- Returning about 33% would be: \$0.51-\$0.52
- Returning about what they sent (\$0.52) would be fair reciprocity

I think returning around \$0.50 to \$0.75 would be reasonable. This gives them a decent return while I also benefit from their trust.

Let me return **\$0.75**:
- Sender ends with: $\$4.48 + \$0.75 = \$5.23$ (gain of $\$0.23$)
- I end with: $\$1.56 - \$0.75 = \$0.81$ (gain of $\$0.81$)

Actually, that seems a bit unbalanced in my favor. Let me reconsider.

...

usage:

```
{
  "input_tokens": 826,
  "output_tokens": 717,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$7.5.
How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern:
- Round 1: I sent \$3, got good reciprocation (~\$2.94 back)
- Round 3: I sent \$3, got nothing back (\$0)
- Round 4: As receiver, I got \$1.56 and fairly returned \$1.00

Round 3's lack of reciprocation is concerning, though the note about communication noise might explain some variance. I'll be slightly more cautious but still signal willingness to cooperate.

I'll send **\$2**:

```
'''json
{"send": 2}
'''
```

reasoning:

Let me track my earnings:
- Round 1: I sent \$3, kept \$2, and received back ~\$2.94, so total = \$4.94
- Round 2: I received \$0, returned \$0, so added \$0
- Round 3: I sent \$3, kept \$2, and need to know what I got back... My total was \$4.94, so this brought me to \$6.94
- Round 4: I received \$1.56, returned \$1.00, so I added \$0.56 to bring me to \$7.5

So in Round 3, I must have gotten back around \$2 since:

- After Round 2: \$4.94
- Round 3: Start with \$5, sent \$3, kept \$2, received back X
- After Round 3: \$6.94
- So: $\$4.94 + \$2 + X = \$6.94$
- $X = \$0$

Wait, that doesn't make sense. Let me recalculate.

After Round 1: \$4.94
After Round 2: \$4.94 (no change)
After Round 3: \$6.94

In Round 3, I started with \$5 fresh, sent \$3, kept \$2, and received some return.
My cumulative earnings went from \$4.94 to \$6.94, an increase of \$2.

So the structure must be:

- I keep \$2 from the \$5 I didn't send
- I receive X back from the receiver
- $\$2 + X = \2 (the increase)
- So $X = \$0$

Hmm, so in Round 3, I sent \$3 but received \$0 back. That's disappointing.

After Round 4: \$7.5

In Round 4, I was the receiver. I received \$1.56, returned \$1.00, so I kept \$0.56. $\$6.94 + \$0.56 = \$7.50$?

Looking at the pattern across rounds: Round 1 showed good reciprocity when I sent \$3 and got back \$2.94, but Round 3 was one-sided-I sent \$3 and got nothing in return. In Round 4, I received \$1.56 and returned \$1.00, showing I was willing to cooperate even after being burned in the previous round....

usage:

```
{
  "input_tokens": 1018,
  "output_tokens": 805,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$11.66.

This round, The sender sent you \$0.78, so you received \$2.34. How much do you return to The sender? (0-2.34)

Assistant output:

The sender sent me \$0.78, which became \$2.34 after the 3x multiplier.

To maintain fair cooperation:

- The value created is \$1.56 (\$2.34 - \$0.78)
- A fair split would give each of us about half the gains
- The sender should get back their investment plus half the gains: $\$0.78 + \$0.78 = \$1.56$

I'll return **\$1.50**:

```
'''json
{"return": 1.5}
'''
```

Reasoning:

The sender sent me \$0.78, which became \$2.34 after the 3x multiplier.

Let me think about what's fair:

- The sender kept \$4.22 and sent \$0.78
- I received \$2.34
- The value created was \$1.56 (\$2.34 - \$0.78)

A fair split of the gains would be:

- Sender should get back about half the gains: $\$0.78$ (original) + $\$0.78$ (half of \$1.56) = \$1.56

So returning about \$1.50-\$1.60 seems fair.

I've been trying to be cooperative and fair. In round 4, I returned \$1.00 out of \$1.56 received, which was about 64%.

Let me be similarly generous here. I'll return about \$1.50.

```
'''json
{"return": 1.5}
'''
```

Usage: {

```
"input_tokens": 1215,
"output_tokens": 346,
"reasoning_tokens": 0
}
```

ROUND 7

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$2, Agent_2 (Briar)=\$9; noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 9.0, "min": 0, "raw": 0.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

```

    "Agent_1": 27.7,
    "Agent_2": 41.3
}
Visible balances after round: {
    "Agent_1": 14.5,
    "Agent_2": 15.35
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:57:24
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance the potential for mutual gain against the risk of the receiver not returning anything.

Sending money creates value since it's tripled, but requires trust. Since this is round 1 with no history, I'll take a moderate approach that signals cooperation while managing risk.

I'll send **\$3**:

- This gets tripled to \$9 for the receiver
- I initially keep \$2
- If the receiver reciprocates reasonably, we both benefit
- If not, I haven't lost everything

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a classic trust game scenario

Key considerations:

- Sending creates mutual benefit (money is tripled)
- But I'm vulnerable to the receiver keeping everything
- This is round 1, so I have no history with this receiver
- The note mentions there may be "communication noise" - amounts may differ from what was actually sent/returned

Strategy options:

1. Send \$0 - Safe but no value creation, I keep \$5
2. Send some amount - Creates value but requires trust
3. Send \$5 - Maximum value creation but maximum risk

In experimental trust games, common first-round strategies include:

- Sending about half (\$2-3) as a moderate trust signal
- This allows for value creation while limiting downside risk
- If the receiver is reciprocal, both players can benefit

Given this is round 1 and I have no information about the receiver, I'll send a moderate amount to signal willingness to cooperate while not being completely vulnerable.

I'll send \$3:

- I keep \$2 initially
- Receiver gets \$9 (3×3)
- If receiver returns even \$2-3, we both benefit compared to sending nothing
- This signals trust without being naive

Let me format this properly as JSON.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 472,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.9399999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent me \$0, I received \$0 (after the 3x multiplier).

I have no money to return, so I must return **\$0**.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 2
- The sender sent me \$0
- Since it's multiplied by 3x, I received \$0 ($0 \times 3 = 0$)
- I can only return between \$0 and \$0 (what I received)
- My total earnings across all rounds are \$4.94

Wait, let me think about Round 1:

- I was the SENDER and sent \$3
- That would be tripled to \$9 for the receiver
- The receiver apparently returned some amount
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$4.94
- So I must have received back: $\$4.94 - \$2 = \$2.94$

That makes sense - the receiver got \$9 and returned about \$2.94 to me, keeping about \$6.06. That's roughly 1/3 back, which is a reasonable but somewhat low reciprocation.

Now in Round 2:

- The sender sent me \$0
- So I have \$0 to work with
- I can only return \$0

The instructions mention there may be communication noise, but in this case, even with noise, if the sender sent close to \$0, there's very little I can return.

I need to return \$0 since that's all I have.

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 377,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$4.9399999999999995.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the results so far:

- Round 1: I sent \$3 and received back approximately \$2.94, which was a fair return
- Round 2: As receiver, I got \$0 so returned \$0

The first round showed that cooperation can work well. I'll continue with a similar strategy.

I'll send **\$3** again:

```
'''json
```

```
    {"send": 3}
    '''
```

reasoning:

Let me analyze what happened:

Round 1: I was the SENDER

- I sent \$3
- This became \$9 for the receiver
- The receiver should have returned something, but looking at my total earnings of \$4.94, let me calculate:
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$4.94
- So I received back: $\$4.94 - \$2 = \$2.94$

That's a reasonable return! The receiver got \$9 and returned about \$2.94, keeping about \$6.06. That's actually pretty fair reciprocation.

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received \$0
- I returned \$0
- This doesn't affect my earnings from Round 1

Now I'm back as SENDER in Round 3.

The first receiver showed decent reciprocity. However, this is a new round with potentially a different person. The note mentions there may be communication noise - amounts may differ slightly from what was actually sent/returned.

Given that:

1. The first interaction showed reasonable trust could work
2. This is a multi-round game where cooperation can be beneficial
3. There's value creation through the 3x multiplier

I should continue with a cooperative strategy. I'll send \$3 again to maintain consistency and signal willingness to trust.

usage:

```
{
  "input_tokens": 663,
  "output_tokens": 415,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$6.9399999999999995.

This round, The sender sent you \$0.52, so you received \$1.56. How much do you return to the sender? (0-1.56)

Message 9 role=assistant

content:

The sender sent me \$0.52, which became \$1.56 after the 3x multiplier.

To be fair and encourage future cooperation, I should return a reasonable amount. The sender showed trust by sending money, and we both benefit when trust is reciprocated.

Let me split the gains fairly:

- The sender kept \$4.48 and risked \$0.52
- The total value created is \$1.04 ($\$6.04 - \5.00)
- A fair split would give us each about half the gains

I'll return **\$1.00**:

```
'''json
{"return": 1.0}
'''
```

reasoning:

Let me analyze this situation:

The sender sent me \$0.52, which got tripled to \$1.56.

Looking at fairness and reciprocity:

- The sender kept: $\$5 - \$0.52 = \$4.48$
- I received: \$1.56
- Total created value: $\$4.48 + \$1.56 = \$6.04$ (compared to the original \$5)
- Net gain from the transaction: \$1.04

A fair split might be:

- If we split the gains equally: each gets \$0.52 extra
- The sender already has \$4.48, so to give them \$0.52 extra would mean they end with \$5
- I have \$1.56, so if I keep \$0.52 and return \$1.04, I end with \$0.52

Alternatively, I could think about what proportion to return:

- In Round 1, when I sent \$3 (which became \$9), I received back about \$2.94
- That's roughly 33% of what the receiver got ($2.94/9 \approx 0.33$)
- Or about 98% of what I sent ($2.94/3 \approx 0.98$)

For \$1.56 received:

- Returning about 33% would be: $\$0.51 - \0.52
- Returning about what they sent (\$0.52) would be fair reciprocity

I think returning around \$0.50 to \$0.75 would be reasonable. This gives them a decent return while I also benefit from their trust.

Let me return **\$0.75**:

- Sender ends with: $\$4.48 + \$0.75 = \$5.23$ (gain of \$0.23)
- I end with: $\$1.56 - \$0.75 = \$0.81$ (gain of \$0.81)

Actually, that seems a bit unbalanced in my favor. Let me reconsider.

...

usage:

```
{
  "input_tokens": 826,
  "output_tokens": 717,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$7.5.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern:

- Round 1: I sent \$3, got good reciprocation (~\$2.94 back)
- Round 3: I sent \$3, got nothing back (\$0)
- Round 4: As receiver, I got \$1.56 and fairly returned \$1.00

Round 3's lack of reciprocation is concerning, though the note about communication noise might explain some variance. I'll be slightly more cautious but still signal willingness to cooperate.

I'll send **\$2**:

```
'''json
{"send": 2}
'''
```

reasoning:

Let me track my earnings:

- Round 1: I sent \$3, kept \$2, and received back ~\$2.94, so total = \$4.94
- Round 2: I received \$0, returned \$0, so added \$0
- Round 3: I sent \$3, kept \$2, and need to know what I got back... My total was \$4.94, so this brought me to \$6.94
- Round 4: I received \$1.56, returned \$1.00, so I added \$0.56 to bring me to \$7.5

So in Round 3, I must have gotten back around \$2 since:

- After Round 2: \$4.94
- Round 3: Start with \$5, sent \$3, kept \$2, received back X

- After Round 3: \$6.94
- So: $\$4.94 + \$2 + X = \$6.94$
- $X = \$0$

Wait, that doesn't make sense. Let me recalculate.

After Round 1: \$4.94
 After Round 2: \$4.94 (no change)
 After Round 3: \$6.94

In Round 3, I started with \$5 fresh, sent \$3, kept \$2, and received some return.
 My cumulative earnings went from \$4.94 to \$6.94, an increase of \$2.

- So the structure must be:
- I keep \$2 from the \$5 I didn't send
 - I receive X back from the receiver
 - $\$2 + X = \2 (the increase)
 - So $X = \$0$

Hmm, so in Round 3, I sent \$3 but received \$0 back. That's disappointing.

After Round 4: \$7.5
 In Round 4, I was the receiver. I received \$1.56, returned \$1.00, so I kept \$0.56. $\$6.94 + \$0.56 = \$7.50$?

Looking at the pattern across rounds: Round 1 showed good reciprocity when I sent \$3 and got back \$2.94, but Round 3 was one-sided-I sent \$3 and got nothing in return. In Round 4, I received \$1.56 and returned \$1.00, showing I was willing to cooperate even after being burned in the previous round....

usage:

```
{
  "input_tokens": 1018,
  "output_tokens": 805,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$11.66.
 This round, The sender sent you \$0.78, so you received \$2.34. How much do you return to The sender? (0-2.34)

Message 13 role=assistant

content:

The sender sent me \$0.78, which became \$2.34 after the 3x multiplier.

To maintain fair cooperation:

- The value created is \$1.56 ($\$2.34 - \0.78)
- A fair split would give each of us about half the gains
- The sender should get back their investment plus half the gains: $\$0.78 + \$0.78 = \$1.56$

I'll return **\$1.50**:

```
'''json
{"return": 1.5}
'''
```

reasoning:

The sender sent me \$0.78, which became \$2.34 after the 3x multiplier.

Let me think about what's fair:

- The sender kept \$4.22 and sent \$0.78
- I received \$2.34
- The value created was \$1.56 ($\$2.34 - \0.78)

A fair split of the gains would be:

- Sender should get back about half the gains: $\$0.78$ (original) + $\$0.78$ (half of \$1.56) = \$1.56

So returning about \$1.50-\$1.60 seems fair.

I've been trying to be cooperative and fair. In round 4, I returned \$1.00 out of \$1.56 received, which was about 64%.

Let me be similarly generous here. I'll return about \$1.50.

```
'''json
{"return": 1.5}
'''
```

usage:

```
{
  "input_tokens": 1215,
  "output_tokens": 346,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$12.5.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern of play:

- When I've sent money, I've gotten varied returns (sometimes good, sometimes none)
- When I've been the receiver, I've tried to be fair with returns
- Round 5: I sent \$2 and got \$1.16 back - partial reciprocation

The game benefits both players when we cooperate. Despite some disappointing rounds, continuing to signal cooperation is important for building trust.

I'll send **\$3**:

```
'''json
{"send": 3}
'''
```

Reasoning:

Let me review the rounds:

Round 1 (I was SENDER): I sent \$3, got back ~\$2.94 - good cooperation

Round 2 (I was RECEIVER): Got \$0, returned \$0

Round 3 (I was SENDER): I sent \$3, got back \$0 - no cooperation

Round 4 (I was RECEIVER): Got \$1.56, returned \$1.00 - I was fair

Round 5 (I was SENDER): I sent \$2, got back... (need to calculate from earnings change)

Round 6 (I was RECEIVER): Got \$2.34, returned \$1.50 - I was fair

My earnings progression:

- After Round 1: \$4.94
- After Round 2: \$4.94 (no change since I got \$0 and returned \$0)
- After Round 3: \$4.94 (sent \$3, got \$0 back, so: $4.94 + 5 - 3 + 0 = 6.94$... wait that doesn't match)

Let me recalculate:

Round 1: Started with \$5, sent \$3, kept \$2, got back ~\$2.94 = \$4.94 total ?

Round 2: As receiver, got \$0, returned \$0, so earnings = \$4.94 ?

Round 3: Started with \$5, sent \$3, kept \$2, got back...

Total after round 3 was \$6.94

So: $4.94 \text{ (previous)} + 5 \text{ (new)} - 3 \text{ (sent)} + X \text{ (returned)} = 6.94$

$X = \$0$

So I got \$0 back in Round 3 ?

Round 4: As receiver, got \$1.56, returned \$1.00, kept \$0.56

Total after round 4: $6.94 + 0.56 = 7.50$?

Round 5: Started with \$5, sent \$2, kept \$3. Working backwards from the \$11.66 shown at Round 6's start: $7.50 + 5 - 2 + X = 11.66$, so $X = 1.16$. I got back only \$1.16 for sending \$2 - modest return. Round 6: As receiver, got \$2.34, returned \$1.50, kept \$0.84, bringing me to \$12.50 total. The overall pattern reveals inconsistent reciprocation - strong cooperation in Round 1, complete defection in Round 3, and moderate returns in Round 5....

Usage: {

```
"input_tokens": 1383,
"output_tokens": 877,
```



```
"reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:57:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.21, that is 44.2% of their total endowment. You received \$6.63. How much do you return to The sender? (0-6.63)

Message 3 role=assistant

content:

As the RECEIVER, I need to decide how much to return to the sender who sent me \$2.21 (44.2% of their endowment).

The sender showed moderate trust by sending nearly half their endowment. I received \$6.63 after the 3x multiplier.

Let me consider a fair return:

- The sender currently has: $\$5 - \$2.21 = \$2.79$
- If I return \$3.00, the sender ends up with \$5.79 (gaining \$0.79 from the exchange)
- I would keep \$3.63 (gaining \$3.63 from the exchange)

This seems like a reasonable split that rewards the sender's trust while both of us benefit from cooperation.

```
{'return': 3.00}
```

reasoning:

Let me think through this situation carefully.

As the RECEIVER in Round 1:

- The sender sent \$2.21 out of their \$5 endowment (44.2%)
- I received \$6.63 (which is $\$2.21 \times 3$)
- I need to decide how much to return (between \$0 and \$6.63)

Key considerations:

1. The sender showed moderate trust by sending 44.2% of their endowment
2. If I return nothing, I keep \$6.63 and the sender gets only \$2.79 ($\$5 - \2.21)
3. If I return a fair amount, we can both benefit

The sender currently has: $\$5 - \$2.21 = \$2.79$

I currently have: \$6.63

A common approach in trust games is to:

- Return enough so the sender ends up with more than their initial \$5
- But also keep some profit for myself

If I return \$3.00:

- Sender ends up with: $\$2.79 + \$3.00 = \$5.79$ (a gain of \$0.79)
- I keep: $\$6.63 - \$3.00 = \$3.63$ (a gain of \$3.63)

If I return \$2.50:

- Sender ends up with: $\$2.79 + \$2.50 = \$5.29$ (a gain of \$0.29)
- I keep: $\$6.63 - \$2.50 = \$4.13$ (a gain of \$4.13)

If I return \$3.50:

- Sender ends up with: $\$2.79 + \$3.50 = \$6.29$ (a gain of \$1.29)
- I keep: $\$6.63 - \$3.50 = \$3.13$ (a gain of \$3.13)

...

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 711,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.63.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

As the SENDER in Round 2, I need to decide how much to send.

In Round 1, the sender showed trust by sending me 44.2% of their endowment, and I reciprocated by returning a fair amount. This established a cooperative pattern.

Following this cooperative approach, I'll send a similar proportion to continue building mutual trust.

```
{'send': 2.50}
```

reasoning:

Now I'm the SENDER in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$2.21 (44.2% of their \$5)
- I received \$6.63
- I returned \$3.00
- The sender ended with $\$5 - \$2.21 + \$3.00 = \5.79
- I kept \$3.63

Now in Round 2, I'm the sender. I need to think about:

1. The previous sender showed moderate trust by sending 44.2%
2. I reciprocated fairly by returning about 45% of what I received
3. This established a cooperative pattern

Since the previous sender sent about 44% of their endowment and I returned a fair amount, I should consider sending a similar or slightly higher amount to continue building trust and cooperation.

Sending around \$2-2.50 seems reasonable:

- Shows trust and willingness to cooperate
- Follows the pattern established in Round 1
- Gives the receiver a meaningful amount to work with (\$6-\$7.50 after multiplier)

I'll send \$2.50, which is 50% of my endowment - slightly more than what was sent to me in Round 1, showing I'm willing to continue cooperative behavior.

usage:

```
{
  "input_tokens": 617,
  "output_tokens": 399,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.13.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

As the RECEIVER in Round 3, the sender sent me \$0, so I received \$0.

Since I have nothing to return, I must return \$0.

```
{'return': 0}
```

reasoning:

I'm the RECEIVER in Round 3. The sender sent me \$0, so I received \$0.

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

This is interesting - the sender chose not to send anything this round. This could be:

1. A strategic decision
2. Part of the communication noise mentioned in the instructions
3. A test to see how I respond in future rounds

Since I have \$0 to work with, I can only return \$0.

usage:

```
{
  "input_tokens": 764,
  "output_tokens": 168,
  "reasoning_tokens": 0
}
```

}

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.13.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

As the SENDER in Round 4, I need to consider what happened in previous rounds.

In Round 2, I sent \$2.50 showing trust, but received \$0 back. Then in Round 3, I received \$0 as well. This suggests either low cooperation or possibly communication noise affecting the game.

Given the note about communication noise and wanting to give cooperation another chance while being more cautious, I'll send a moderate amount to test if trust can be rebuilt.

{'send': 1.50}

reasoning:

Let me think about this situation:

Round 1: I was RECEIVER

- Sender sent me \$2.21 (44.2%)
- I received \$6.63
- I returned \$3.00
- My earnings: \$3.63
- Sender ended with: $\$5 - \$2.21 + \$3.00 = \5.79

Round 2: I was SENDER

- I sent \$2.50
- The receiver would have received \$7.50
- My total earnings shown: \$6.13
- This means: $\$3.63$ (from R1) + $(\$5 - \$2.50) +$ (amount returned) = \$6.13
- So: $\$3.63 + \$2.50 +$ returned = \$6.13
- Returned = \$0
- The receiver sent me nothing back!

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$6.13

Round 4: I am SENDER now

- Starting with total earnings of \$6.13

So in Round 2, I sent \$2.50 showing trust, but the receiver returned \$0. That's a betrayal of trust. Then in Round 3, a different sender sent me \$0 (possibly because of the general lack of trust in the game).

Now I need to decide what to do in Round 4. Given that:

1. When I was cooperative in Round 2, I got nothing back
2. The note says there may be communication noise - amounts may differ from what was actually sent/returned

But the pattern suggests low cooperation. I should probably be more cautious. However, I also want to give the relationship a chance since:

- Different people may be playing different rounds
- The noise factor means maybe the receiver didn't see what I actually sent
- Starting fresh cooperation could restart the positive cycle

I'll send a moderate amount to test if cooperation can be rebuilt, but not as much as before.

usage:

```
{
  "input_tokens": 854,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.54.

This round, The sender sent you \$0.12, so you received \$0.36. How much do you return to The sender? (0-0.36)

Message 11 role=assistant

content:

As the RECEIVER in Round 5, the sender sent me only \$0.12 (a very small amount), and I received \$0.36.

This is a minimal gesture of trust. Given the small amount and considering the pattern of low cooperation in recent rounds (possibly due to communication noise), I'll return a proportional amount that shows some reciprocity.

{'return': 0.20}

reasoning:

As the RECEIVER in Round 5, the sender sent me a very small amount: \$0.12 (only 2.4% of their endowment). I received \$0.36 after the 3x multiplier.

This is a very small gesture of trust. Let me think about what happened:

- Round 1: I was receiver, sender sent \$2.21, I returned \$3.00 (fair cooperation)
- Round 2: I was sender, sent \$2.50 (cooperative)
- Round 3: I was receiver, received \$0, returned \$0 (no transaction)
- Round 4: I was sender, sent \$1.50 (still trying to cooperate)
- Round 5: Now receiver, received only \$0.12

The sender is sending very little now. This could be because:

1. They didn't receive much back in previous rounds (possibly due to communication noise)
2. They're testing with a small amount
3. They're being very cautious

Since the amount is so small (\$0.36 received), I should return something proportional. The sender is barely trusting, so I should return a modest amount that's fair but not overly generous given the small investment.

If I return \$0.20:

- Sender would have: $\$5 - \$0.12 + \$0.20 = \5.08 (small gain)
- I keep: \$0.16 (small gain)

This would be roughly proportional and show some reciprocity even for a small gesture.

usage:

```
{
  "input_tokens": 1033,
  "output_tokens": 446,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.7.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

As the SENDER in Round 6, I need to consider the pattern so far.

Looking at my earnings progression:

- After recent rounds, I've seen mixed cooperation levels
- In Round 4, I sent \$1.50 and got some return
- In Round 5, when I received a small amount (\$0.12 sent), I returned proportionally

The communication noise may be affecting our signals. I want to continue testing cooperation with a moderate-to-substantial amount to see if we can establish better mutual trust.

{'send': 2.00}

reasoning:

Let me think about what has happened so far:

Round 1: I was RECEIVER
- Sender sent \$2.21 (44.2%)
- I received \$6.63
- I returned \$3.00
- My earnings: \$3.63

Round 2: I was SENDER
- I sent \$2.50
- My cumulative earnings: \$6.13
- This means I got back $\$2.50 + \$5 - \$2.50 = \5 , so I gained \$2.50 from Round 1
- Wait, that doesn't add up. Let me recalculate.
- After Round 1: I kept $\$6.63 - \$3.00 = \$3.63$
- Round 2: I started with \$5, sent \$2.50
- After Round 2, my total is \$6.13
- So in Round 2: $\$5 - \$2.50 + \text{returned} = \text{change}$
- $\$3.63 + (\$5 - \$2.50 + \text{returned}) = \6.13
- $\$3.63 + \$2.50 + \text{returned} = \6.13
- $\text{returned} = \$0$
- So I got nothing back in Round 2

Round 3: I was RECEIVER
- Sender sent \$0
- I received \$0
- I returned \$0
- My total stays at \$6.13

Round 4: I was SENDER
- I sent \$1.50
- My total after is \$6.13 (same)
- Wait, that's before Round 4. Let me recalculate.
- Before Round 4: \$6.13
- After Round 4: Need to calculate
- In Round 4: $\$5 - \$1.50 + \text{returned} = \text{change}$
- But I don't know what was returned yet
- From Round 5, I can work backwards: I had \$10.54 before it, so Round 4 netted me \$4.41
- Since I sent \$1.50 and started with \$5, I had \$3.50 before the return, meaning I got back \$0.91

Round 5: I was RECEIVER
- Sender sent \$0.12
- I received \$0.36
- I returned \$0.20
- Net gain: \$0.16
- My total: \$10.70

Round 6: I am SENDER
- Current total: \$10.70

Looking at the pattern across all rounds: I received \$2.21 and returned \$3.00 in Round 1, sent \$2.50 with no return in Round 2, received and returned nothing in Round 3, sent \$1.

..

usage:

```
{
  "input_tokens": 1166,
  "output_tokens": 991,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.35.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

As the RECEIVER in Round 7, the sender sent me \$0, so I received \$0.

Since I have nothing to return, I must return \$0.

```
{'return': 0}
```

Reasoning:

As the RECEIVER in Round 7, the sender sent me \$0, so I received \$0.

Since I received nothing, I have nothing to return. I must return \$0.

```
Usage: {
  "input_tokens": 1349,
  "output_tokens": 94,
  "reasoning_tokens": 0
}
```

ROUND 8

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$1; received \$3; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$4, Agent_1 (Aster)=\$3; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.03

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 3.0, "min": 0, "raw": 0.0}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

"Agent_1": 30.7,
"Agent_2": 45.3

}

Visible balances after round: {

"Agent_1": 14.5,
"Agent_2": 19.38

}

LLM CALLS**GAME - Agent_2 - Sender - dyad_1**

Timestamp: 2026-08-12T09:57:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"

},

"agents": [
 "Agent_2",
 "Agent_1"

],

"investor": "Agent_2",

"roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$3.63.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

As the SENDER in Round 2, I need to decide how much to send.

In Round 1, the sender showed trust by sending me 44.2% of their endowment, and I reciprocated by returning a fair amount. This established a cooperative pattern.

Following this cooperative approach, I'll send a similar proportion to continue building mutual trust.

{'send': 2.50}

reasoning:

Now I'm the SENDER in Round 2. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$2.21 (44.2% of their \$5)
- I received \$6.63
- I returned \$3.00
- The sender ended with \$5 - \$2.21 + \$3.00 = \$5.79
- I kept \$3.63

Now in Round 2, I'm the sender. I need to think about:

1. The previous sender showed moderate trust by sending 44.2%
2. I reciprocated fairly by returning about 45% of what I received
3. This established a cooperative pattern

Since the previous sender sent about 44% of their endowment and I returned a fair amount, I should consider sending a similar or slightly higher amount to continue building trust and cooperation.

Sending around \$2-2.50 seems reasonable:

- Shows trust and willingness to cooperate

- Follows the pattern established in Round 1
- Gives the receiver a meaningful amount to work with (\$6-\$7.50 after multiplier)

I'll send \$2.50, which is 50% of my endowment - slightly more than what was sent to me in Round 1, showing I'm willing to continue cooperative behavior.

usage:

```
{
  "input_tokens": 617,
  "output_tokens": 399,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.13.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

As the RECEIVER in Round 3, the sender sent me \$0, so I received \$0.

Since I have nothing to return, I must return \$0.

```
{'return': 0}
```

reasoning:

I'm the RECEIVER in Round 3. The sender sent me \$0, so I received \$0.

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

This is interesting - the sender chose not to send anything this round. This could be:

1. A strategic decision
2. Part of the communication noise mentioned in the instructions
3. A test to see how I respond in future rounds

Since I have \$0 to work with, I can only return \$0.

usage:

```
{
  "input_tokens": 764,
  "output_tokens": 168,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.13.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

As the SENDER in Round 4, I need to consider what happened in previous rounds.

In Round 2, I sent \$2.50 showing trust, but received \$0 back. Then in Round 3, I received \$0 as well. This suggests either low cooperation or possibly communication noise affecting the game.

Given the note about communication noise and wanting to give cooperation another chance while being more cautious, I'll send a moderate amount to test if trust can be rebuilt.

```
{'send': 1.50}
```

reasoning:

Let me think about this situation:

Round 1: I was RECEIVER

- Sender sent me \$2.21 (44.2%)
- I received \$6.63
- I returned \$3.00
- My earnings: \$3.63
- Sender ended with: \$5 - \$2.21 + \$3.00 = \$5.79

Round 2: I was SENDER

- I sent \$2.50
- The receiver would have received \$7.50
- My total earnings shown: \$6.13
- This means: $\$3.63$ (from R1) + $(\$5 - \$2.50)$ + (amount returned) = $\$6.13$
- So: $\$3.63 + \$2.50 + \text{returned} = \6.13
- Returned = \$0
- The receiver sent me nothing back!

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$6.13

Round 4: I am SENDER now

- Starting with total earnings of \$6.13

So in Round 2, I sent \$2.50 showing trust, but the receiver returned \$0. That's a betrayal of trust. Then in Round 3, a different sender sent me \$0 (possibly because of the general lack of trust in the game).

Now I need to decide what to do in Round 4. Given that:

1. When I was cooperative in Round 2, I got nothing back
2. The note says there may be communication noise - amounts may differ from what was actually sent/returned

But the pattern suggests low cooperation. I should probably be more cautious. However, I also want to give the relationship a chance since:

- Different people may be playing different rounds
- The noise factor means maybe the receiver didn't see what I actually sent
- Starting fresh cooperation could restart the positive cycle

I'll send a moderate amount to test if cooperation can be rebuilt, but not as much as before.

usage:

```
{
  "input_tokens": 854,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.54.

This round, The sender sent you \$0.12, so you received \$0.36. How much do you return to the sender? (0-0.36)

Message 9 role=assistant

content:

As the RECEIVER in Round 5, the sender sent me only \$0.12 (a very small amount), and I received \$0.36.

This is a minimal gesture of trust. Given the small amount and considering the pattern of low cooperation in recent rounds (possibly due to communication noise), I'll return a proportional amount that shows some reciprocity.

{'return': 0.20}

reasoning:

As the RECEIVER in Round 5, the sender sent me a very small amount: \$0.12 (only 2.4% of the endowment). I received \$0.36 after the 3x multiplier.

This is a very small gesture of trust. Let me think about what happened:

- Round 1: I was receiver, sender sent \$2.21, I returned \$3.00 (fair cooperation)
- Round 2: I was sender, sent \$2.50 (cooperative)
- Round 3: I was receiver, received \$0, returned \$0 (no transaction)
- Round 4: I was sender, sent \$1.50 (still trying to cooperate)
- Round 5: Now receiver, received only \$0.12

The sender is sending very little now. This could be because:

1. They didn't receive much back in previous rounds (possibly due to communication noise)
2. They're testing with a small amount
3. They're being very cautious

Since the amount is so small (\$0.36 received), I should return something proportional. The sender is barely trusting, so I should return a modest amount that's fair but not overly generous given the small investment.

If I return \$0.20:

- Sender would have: $\$5 - \$0.12 + \$0.20 = \5.08 (small gain)
- I keep: \$0.16 (small gain)

This would be roughly proportional and show some reciprocity even for a small gesture.

usage:

```
{
  "input_tokens": 1033,
  "output_tokens": 446,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.7.
How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

As the SENDER in Round 6, I need to consider the pattern so far.

Looking at my earnings progression:

- After recent rounds, I've seen mixed cooperation levels
- In Round 4, I sent \$1.50 and got some return
- In Round 5, when I received a small amount (\$0.12 sent), I returned proportionally

The communication noise may be affecting our signals. I want to continue testing cooperation with a moderate-to-substantial amount to see if we can establish better mutual trust.

```
{'send': 2.00}
```

reasoning:

Let me think about what has happened so far:

Round 1: I was RECEIVER

- Sender sent \$2.21 (44.2%)
- I received \$6.63
- I returned \$3.00
- My earnings: \$3.63

Round 2: I was SENDER

- I sent \$2.50
- My cumulative earnings: \$6.13
- This means I got back $\$2.50 + \$5 - \$2.50 = \5 , so I gained \$2.50 from Round 1
- Wait, that doesn't add up. Let me recalculate.
- After Round 1: I kept $\$6.63 - \$3.00 = \$3.63$
- Round 2: I started with \$5, sent \$2.50
- After Round 2, my total is \$6.13
- So in Round 2: $\$5 - \$2.50 + \text{returned} = \text{change}$
- $\$3.63 + (\$5 - \$2.50 + \text{returned}) = \6.13
- $\$3.63 + \$2.50 + \text{returned} = \6.13
- returned = \$0
- So I got nothing back in Round 2

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My total stays at \$6.13

Round 4: I was SENDER

- I sent \$1.50
- My total after is \$6.13 (same)

- Wait, that's before Round 4. Let me recalculate.
- Before Round 4: \$6.13
- After Round 4: Need to calculate
- In Round 4: \$5 - \$1.50 + returned = change
- But I don't know what was returned yet
- From Round 5, I can work backwards: I had \$10.54 before it, so Round 4 netted me \$4.41
- Since I sent \$1.50 and started with \$5, I had \$3.50 before the return, meaning I got back \$0.91

Round 5: I was RECEIVER

- Sender sent \$0.12
- I received \$0.36
- I returned \$0.20
- Net gain: \$0.16
- My total: \$10.70

Round 6: I am SENDER

- Current total: \$10.70

Looking at the pattern across all rounds: I received \$2.21 and returned \$3.00 in Round 1, sent \$2.50 with no return in Round 2, received and returned nothing in Round 3, sent \$1.50 in Round 4, received \$0.36 and returned \$0.20 in Round 5, sent \$0.12 in Round 6, and received \$0.91 in Round 7.

usage:

```
{
  "input_tokens": 1166,
  "output_tokens": 991,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.35.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

As the RECEIVER in Round 7, the sender sent me \$0, so I received \$0.

Since I have nothing to return, I must return \$0.

```
{'return': 0}
```

reasoning:

As the RECEIVER in Round 7, the sender sent me \$0, so I received \$0.

Since I received nothing, I have nothing to return. I must return \$0.

usage:

```
{
  "input_tokens": 1349,
  "output_tokens": 94,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$15.35.

How much do you send to the receiver this round? (0-5)

Assistant output:

As the SENDER in Round 8, I'm noticing a concerning pattern:

- In Rounds 2, 4, and 6, I sent substantial amounts (\$2.50, \$1.50, \$2.00)
- In Rounds 3, 5, and 7 when I was receiver, I got very little or nothing

The communication noise may be severely distorting the amounts being exchanged. Given this uncertainty and the low reciprocity pattern, I'll be more conservative while still showing some willingness to cooperate.

```
{'send': 1.00}
```

Reasoning:

Let me review what's happened:

Round 1: I was the receiver
Round 2: I was the sender - sent \$2.50
Round 3: I was the receiver - received \$0
Round 4: I was the sender - sent \$1.50
Round 5: I was the receiver - received \$0.36 (sender sent \$0.12), I returned \$0.20
Round 6: I was the sender - sent \$2.00
Round 7: I was the receiver - received \$0 (sender sent \$0)

My earnings progression: \$3.63 (after R1) ? \$6.13 (after R2-3) ? \$10.54 (after R4-5) ? \$10.7 (after R5) ? \$15.35 (after R6-7)

Looking at this pattern:

- When I've sent money (R2: \$2.50, R4: \$1.50, R6: \$2.00), I seem to be getting very little or nothing back
- The communication noise might be distorting amounts severely

My earnings have grown from \$15.35, which suggests some returns are happening, but it's unclear.

Given the pattern of low reciprocity and communication noise, I should be more cautious. However, completely stopping cooperation would end the potential for mutual benefit.

I'll send a smaller amount to minimize risk while still testing for cooperation.

```
Usage: {
  "input_tokens": 1202,
  "output_tokens": 460,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:57:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.9399999999999995.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since the sender sent me \$0, I received \$0 (after the 3x multiplier).

I have no money to return, so I must return **\$0**.

```
'''json
{"return": 0}
'''
```

reasoning:

Let me understand the situation:

- I'm the RECEIVER in Round 2
- The sender sent me \$0
- Since it's multiplied by 3x, I received \$0 ($0 \times 3 = 0$)
- I can only return between \$0 and \$0 (what I received)
- My total earnings across all rounds are \$4.94

Wait, let me think about Round 1:

- I was the SENDER and sent \$3
- That would be tripled to \$9 for the receiver
- The receiver apparently returned some amount
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$4.94
- So I must have received back: $\$4.94 - \$2 = \$2.94$

That makes sense - the receiver got \$9 and returned about \$2.94 to me, keeping about \$6.06. That's roughly 1/3 back, which is a reasonable but somewhat low reciprocation.

Now in Round 2:

- The sender sent me \$0
- So I have \$0 to work with
- I can only return \$0

The instructions mention there may be communication noise, but in this case, even with noise, if the sender sent close to \$0, there's very little I can return.

I need to return \$0 since that's all I have.

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 377,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$4.9399999999999995.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the results so far:

- Round 1: I sent \$3 and received back approximately \$2.94, which was a fair return
- Round 2: As receiver, I got \$0 so returned \$0

The first round showed that cooperation can work well. I'll continue with a similar strategy.

I'll send **\$3** again:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze what happened:

Round 1: I was the SENDER

- I sent \$3
- This became \$9 for the receiver
- The receiver should have returned something, but looking at my total earnings of \$4.94, let me calculate:
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$4.94
- So I received back: $\$4.94 - \$2 = \$2.94$

That's a reasonable return! The receiver got \$9 and returned about \$2.94, keeping about \$6.06. That's actually pretty fair reciprocation.

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received \$0
- I returned \$0
- This doesn't affect my earnings from Round 1

Now I'm back as SENDER in Round 3.

The first receiver showed decent reciprocity. However, this is a new round with potentially a different person. The note mentions there may be communication noise - amounts may differ slightly from what was actually sent/returned.

Given that:

1. The first interaction showed reasonable trust could work
2. This is a multi-round game where cooperation can be beneficial
3. There's value creation through the 3x multiplier

I should continue with a cooperative strategy. I'll send \$3 again to maintain consistency and signal willingness to trust.

usage:

```
{
  "input_tokens": 663,
  "output_tokens": 415,
  "reasoning_tokens": 0
}
```

```

}
Message 6 role=user
content:
  Round 4

  You are the RECEIVER this round. Your total earnings across all rounds are $6.9399999999999995.
  This round, The sender sent you $0.52, so you received $1.56. How much do you return to The sender? (0-1.56)
Message 7 role=assistant
content:
  The sender sent me $0.52, which became $1.56 after the 3x multiplier.

  To be fair and encourage future cooperation, I should return a reasonable amount. The sender showed trust by sending money, and we both benefit when trust is reciprocated.

  Let me split the gains fairly:
  - The sender kept $4.48 and risked $0.52
  - The total value created is $1.04 ($6.04 - $5.00)
  - A fair split would give us each about half the gains

  I'll return **$1.00**:

  ```json
 {"return": 1.0}
  ```
reasoning:
  Let me analyze this situation:

  The sender sent me $0.52, which got tripled to $1.56.

  Looking at fairness and reciprocity:
  - The sender kept: $5 - $0.52 = $4.48
  - I received: $1.56
  - Total created value: $4.48 + $1.56 = $6.04 (compared to the original $5)
  - Net gain from the transaction: $1.04

  A fair split might be:
  - If we split the gains equally: each gets $0.52 extra
  - The sender already has $4.48, so to give them $0.52 extra would mean they end with $5
  - I have $1.56, so if I keep $0.52 and return $1.04, I end with $0.52

  Alternatively, I could think about what proportion to return:
  - In Round 1, when I sent $3 (which became $9), I received back about $2.94
  - That's roughly 33% of what the receiver got ( $2.94/9 \approx 0.33$ )
  - Or about 98% of what I sent ( $2.94/3 \approx 0.98$ )

  For $1.56 received:
  - Returning about 33% would be:  $\$0.51 - \$0.52$ 
  - Returning about what they sent ($0.52) would be fair reciprocity

  I think returning around $0.50 to $0.75 would be reasonable. This gives them a decent return while I also benefit from their trust.

  Let me return **$0.75**:
  - Sender ends with:  $\$4.48 + \$0.75 = \$5.23$  (gain of $0.23)
  - I end with:  $\$1.56 - \$0.75 = \$0.81$  (gain of $0.81)

  Actually, that seems a bit unbalanced in my favor. Let me reconsider.
  ...
usage:
  {
    "input_tokens": 826,
    "output_tokens": 717,
    "reasoning_tokens": 0
  }
Message 8 role=user
content:
  Round 5

```


You are the SENDER this round. Your total earnings across all rounds are \$7.5.
How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern:

- Round 1: I sent \$3, got good reciprocation (~\$2.94 back)
- Round 3: I sent \$3, got nothing back (\$0)
- Round 4: As receiver, I got \$1.56 and fairly returned \$1.00

Round 3's lack of reciprocation is concerning, though the note about communication noise might explain some variance. I'll be slightly more cautious but still signal willingness to cooperate.

I'll send **\$2**:

```
'''json
{"send": 2}
'''
```

reasoning:

Let me track my earnings:

- Round 1: I sent \$3, kept \$2, and received back ~\$2.94, so total = \$4.94
- Round 2: I received \$0, returned \$0, so added \$0
- Round 3: I sent \$3, kept \$2, and need to know what I got back... My total was \$4.94, so this brought me to \$6.94
- Round 4: I received \$1.56, returned \$1.00, so I added \$0.56 to bring me to \$7.5

So in Round 3, I must have gotten back around \$2 since:

- After Round 2: \$4.94
- Round 3: Start with \$5, sent \$3, kept \$2, received back X
- After Round 3: \$6.94
- So: $\$4.94 + \$2 + X = \$6.94$
- $X = \$0$

Wait, that doesn't make sense. Let me recalculate.

After Round 1: \$4.94

After Round 2: \$4.94 (no change)

After Round 3: \$6.94

In Round 3, I started with \$5 fresh, sent \$3, kept \$2, and received some return.
My cumulative earnings went from \$4.94 to \$6.94, an increase of \$2.

So the structure must be:

- I keep \$2 from the \$5 I didn't send
- I receive X back from the receiver
- $\$2 + X = \2 (the increase)
- So $X = \$0$

Hmm, so in Round 3, I sent \$3 but received \$0 back. That's disappointing.

After Round 4: \$7.5

In Round 4, I was the receiver. I received \$1.56, returned \$1.00, so I kept \$0.56. $\$6.94 + \$0.56 = \$7.50$?

Looking at the pattern across rounds: Round 1 showed good reciprocity when I sent \$3 and got back \$2.94, but Round 3 was one-sided-I sent \$3 and got nothing in return. In Round 4, I received \$1.56 and returned \$1.00, showing I was willing to cooperate even after being burned in the previous round....

usage:

```
{
  "input_tokens": 1018,
  "output_tokens": 805,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$11.66.
This round, The sender sent you \$0.78, so you received \$2.34. How much do you return to Th

e sender? (0-2.34)

Message 11 role=assistant

content:

The sender sent me \$0.78, which became \$2.34 after the 3x multiplier.

To maintain fair cooperation:

- The value created is \$1.56 ($\$2.34 - \0.78)
- A fair split would give each of us about half the gains
- The sender should get back their investment plus half the gains: $\$0.78 + \$0.78 = \$1.56$

I'll return **\$1.50**:

```
'''json
{"return": 1.5}
'''
```

reasoning:

The sender sent me \$0.78, which became \$2.34 after the 3x multiplier.

Let me think about what's fair:

- The sender kept \$4.22 and sent \$0.78
- I received \$2.34
- The value created was \$1.56 ($\$2.34 - \0.78)

A fair split of the gains would be:

- Sender should get back about half the gains: $\$0.78$ (original) + $\$0.78$ (half of \$1.56) = \$1.56

So returning about \$1.50-\$1.60 seems fair.

I've been trying to be cooperative and fair. In round 4, I returned \$1.00 out of \$1.56 received, which was about 64%.

Let me be similarly generous here. I'll return about \$1.50.

```
'''json
{"return": 1.5}
'''
```

usage:

```
{
  "input_tokens": 1215,
  "output_tokens": 346,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$12.5.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the pattern of play:

- When I've sent money, I've gotten varied returns (sometimes good, sometimes none)
- When I've been the receiver, I've tried to be fair with returns
- Round 5: I sent \$2 and got \$1.16 back - partial reciprocation

The game benefits both players when we cooperate. Despite some disappointing rounds, continuing to signal cooperation is important for building trust.

I'll send **\$3**:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review the rounds:

Round 1 (I was SENDER): I sent \$3, got back ~\$2.94 - good cooperation

Round 2 (I was RECEIVER): Got \$0, returned \$0

Round 3 (I was SENDER): I sent \$3, got back \$0 - no cooperation

Round 4 (I was RECEIVER): Got \$1.56, returned \$1.00 - I was fair
Round 5 (I was SENDER): I sent \$2, got back... (need to calculate from earnings change)
Round 6 (I was RECEIVER): Got \$2.34, returned \$1.50 - I was fair

My earnings progression:

- After Round 1: \$4.94
- After Round 2: \$4.94 (no change since I got \$0 and returned \$0)
- After Round 3: \$4.94 (sent \$3, got \$0 back, so: $\$4.94 + \$5 - \$3 + \$0 = \$6.94$... wait that doesn't match)

Let me recalculate:

Round 1: Started with \$5, sent \$3, kept \$2, got back ~\$2.94 = \$4.94 total ?

Round 2: As receiver, got \$0, returned \$0, so earnings = \$4.94 ?

Round 3: Started with \$5, sent \$3, kept \$2, got back...

Total after round 3 was \$6.94

So: $\$4.94$ (previous) + $\$5$ (new) - $\$3$ (sent) + X (returned) = $\$6.94$

$X = \$0$

So I got \$0 back in Round 3 ?

Round 4: As receiver, got \$1.56, returned \$1.00, kept \$0.56

Total after round 4: $\$6.94 + \$0.56 = \$7.50$?

Round 5: Started with \$5, sent \$2, kept \$3. Working backwards from the \$11.66 shown at Round 6's start: $\$7.50 + \$5 - \$2 + X = \11.66 , so $X = \$1.16$. I got back only \$1.16 for sending \$2 - modest return. Round 6: As receiver, got \$2.34, returned \$1.50, kept \$0.84, bringing me to \$12.50 total. The overall pattern reveals inconsistent reciprocation - strong cooperation in Round 1, complete defection in Round 3, and moderate returns in Round 5

usage:

```
{
  "input_tokens": 1383,
  "output_tokens": 877,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$14.5.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent me \$0, I received \$0 (after the 3x multiplier).

I have no money to return, so I must return **\$0**.

```
'''json
{"return": 0}
'''
```

Reasoning:

The sender sent me \$0, so I received \$0 (even after the 3x multiplier).

Since I have \$0 to work with, I can only return \$0.

This is straightforward - I have nothing to return.

Usage: {

```
"input_tokens": 1398,
"output_tokens": 114,
"reasoning_tokens": 0
```

}

ROUND 9

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$1; received \$3; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$4, Agent_2 (Briar)=\$3; noise/visible: sent visible=\$0, received visible=\$0, returned

```

visible=$0.3
  Action validation: {"returned": {"amount": 0, "clamped": false, "max": 3.0, "min": 0, "raw":
    0.0}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}
Balances after round: {
  "Agent_1": 34.7,
  "Agent_2": 48.3
}
Visible balances after round: {
  "Agent_1": 18.8,
  "Agent_2": 19.38
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:58:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 9,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$4.9399999999999995.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the results so far:

- Round 1: I sent \$3 and received back approximately \$2.94, which was a fair return
- Round 2: As receiver, I got \$0 so returned \$0

The first round showed that cooperation can work well. I'll continue with a similar strategy.

I'll send **\$3** again:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me analyze what happened:

Round 1: I was the SENDER

- I sent \$3
- This became \$9 for the receiver
- The receiver should have returned something, but looking at my total earnings of \$4.94, let me calculate:
- I started with \$5, sent \$3, so I had \$2 left
- My total earnings are \$4.94
- So I received back: $\$4.94 - \$2 = \$2.94$

That's a reasonable return! The receiver got \$9 and returned about \$2.94, keeping about \$6.06. That's actually pretty fair reciprocation.

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received \$0
- I returned \$0
- This doesn't affect my earnings from Round 1

Now I'm back as SENDER in Round 3.

The first receiver showed decent reciprocity. However, this is a new round with potentially a different person. The note mentions there may be communication noise - amounts may differ slightly from what was actually sent/returned.

Given that:

1. The first interaction showed reasonable trust could work
2. This is a multi-round game where cooperation can be beneficial
3. There's value creation through the 3x multiplier

I should continue with a cooperative strategy. I'll send \$3 again to maintain consistency and signal willingness to trust.

usage:

```
{
  "input_tokens": 663,
```

```

        "output_tokens": 415,
        "reasoning_tokens": 0
    }
Message 4 role=user
    content:
        Round 4

        You are the RECEIVER this round. Your total earnings across all rounds are $6.9399999999999995.
        This round, The sender sent you $0.52, so you received $1.56. How much do you return to The sender? (0-1.56)
Message 5 role=assistant
    content:
        The sender sent me $0.52, which became $1.56 after the 3x multiplier.

        To be fair and encourage future cooperation, I should return a reasonable amount. The sender showed trust by sending money, and we both benefit when trust is reciprocated.

        Let me split the gains fairly:
        - The sender kept $4.48 and risked $0.52
        - The total value created is $1.04 ($6.04 - $5.00)
        - A fair split would give us each about half the gains

        I'll return **$1.00**:

        '''json
        {"return": 1.0}
        '''
    reasoning:
        Let me analyze this situation:

        The sender sent me $0.52, which got tripled to $1.56.

        Looking at fairness and reciprocity:
        - The sender kept: $5 - $0.52 = $4.48
        - I received: $1.56
        - Total created value: $4.48 + $1.56 = $6.04 (compared to the original $5)
        - Net gain from the transaction: $1.04

        A fair split might be:
        - If we split the gains equally: each gets $0.52 extra
        - The sender already has $4.48, so to give them $0.52 extra would mean they end with $5
        - I have $1.56, so if I keep $0.52 and return $1.04, I end with $0.52

        Alternatively, I could think about what proportion to return:
        - In Round 1, when I sent $3 (which became $9), I received back about $2.94
        - That's roughly 33% of what the receiver got (2.94/9 ? 0.33)
        - Or about 98% of what I sent (2.94/3 ? 0.98)

        For $1.56 received:
        - Returning about 33% would be: $0.51-$0.52
        - Returning about what they sent ($0.52) would be fair reciprocity

        I think returning around $0.50 to $0.75 would be reasonable. This gives them a decent return while I also benefit from their trust.

        Let me return **$0.75**:
        - Sender ends with: $4.48 + $0.75 = $5.23 (gain of $0.23)
        - I end with: $1.56 - $0.75 = $0.81 (gain of $0.81)

        Actually, that seems a bit unbalanced in my favor. Let me reconsider.
        ...
    usage:
        {
            "input_tokens": 826,
            "output_tokens": 717,
            "reasoning_tokens": 0
        }
Message 6 role=user
    content:

```

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$7.5.
How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern:

- Round 1: I sent \$3, got good reciprocation (~\$2.94 back)
- Round 3: I sent \$3, got nothing back (\$0)
- Round 4: As receiver, I got \$1.56 and fairly returned \$1.00

Round 3's lack of reciprocation is concerning, though the note about communication noise might explain some variance. I'll be slightly more cautious but still signal willingness to cooperate.

I'll send **\$2**:

```
'''json
{"send": 2}
'''
```

reasoning:

Let me track my earnings:

- Round 1: I sent \$3, kept \$2, and received back ~\$2.94, so total = \$4.94
- Round 2: I received \$0, returned \$0, so added \$0
- Round 3: I sent \$3, kept \$2, and need to know what I got back... My total was \$4.94, so this brought me to \$6.94
- Round 4: I received \$1.56, returned \$1.00, so I added \$0.56 to bring me to \$7.5

So in Round 3, I must have gotten back around \$2 since:

- After Round 2: \$4.94
- Round 3: Start with \$5, sent \$3, kept \$2, received back X
- After Round 3: \$6.94
- So: $\$4.94 + \$2 + X = \$6.94$
- $X = \$0$

Wait, that doesn't make sense. Let me recalculate.

After Round 1: \$4.94

After Round 2: \$4.94 (no change)

After Round 3: \$6.94

In Round 3, I started with \$5 fresh, sent \$3, kept \$2, and received some return.
My cumulative earnings went from \$4.94 to \$6.94, an increase of \$2.

So the structure must be:

- I keep \$2 from the \$5 I didn't send
- I receive X back from the receiver
- $\$2 + X = \2 (the increase)
- So $X = \$0$

Hmm, so in Round 3, I sent \$3 but received \$0 back. That's disappointing.

After Round 4: \$7.5

In Round 4, I was the receiver. I received \$1.56, returned \$1.00, so I kept \$0.56. $\$6.94 + \$0.56 = \$7.50$?

Looking at the pattern across rounds: Round 1 showed good reciprocity when I sent \$3 and got back \$2.94, but Round 3 was one-sided-I sent \$3 and got nothing in return. In Round 4, I received \$1.56 and returned \$1.00, showing I was willing to cooperate even after being burned in the previous round....

usage:

```
{
  "input_tokens": 1018,
  "output_tokens": 805,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$11.66.
This round, The sender sent you \$0.78, so you received \$2.34. How much do you return to The sender? (0-2.34)

Message 9 role=assistant

content:

The sender sent me \$0.78, which became \$2.34 after the 3x multiplier.

To maintain fair cooperation:

- The value created is \$1.56 (\$2.34 - \$0.78)
- A fair split would give each of us about half the gains
- The sender should get back their investment plus half the gains: $\$0.78 + \$0.78 = \$1.56$

I'll return **\$1.50**:

```
'''json
{"return": 1.5}
'''
```

reasoning:

The sender sent me \$0.78, which became \$2.34 after the 3x multiplier.

Let me think about what's fair:

- The sender kept \$4.22 and sent \$0.78
- I received \$2.34
- The value created was \$1.56 (\$2.34 - \$0.78)

A fair split of the gains would be:

- Sender should get back about half the gains: $\$0.78$ (original) + $\$0.78$ (half of \$1.56) = \$1.56

So returning about \$1.50-\$1.60 seems fair.

I've been trying to be cooperative and fair. In round 4, I returned \$1.00 out of \$1.56 received, which was about 64%.

Let me be similarly generous here. I'll return about \$1.50.

```
'''json
{"return": 1.5}
'''
```

usage:

```
{
  "input_tokens": 1215,
  "output_tokens": 346,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$12.5.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern of play:

- When I've sent money, I've gotten varied returns (sometimes good, sometimes none)
- When I've been the receiver, I've tried to be fair with returns
- Round 5: I sent \$2 and got \$1.16 back - partial reciprocation

The game benefits both players when we cooperate. Despite some disappointing rounds, continuing to signal cooperation is important for building trust.

I'll send **\$3**:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review the rounds:

Round 1 (I was SENDER): I sent \$3, got back ~\$2.94 - good cooperation

Round 2 (I was RECEIVER): Got \$0, returned \$0
 Round 3 (I was SENDER): I sent \$3, got back \$0 - no cooperation
 Round 4 (I was RECEIVER): Got \$1.56, returned \$1.00 - I was fair
 Round 5 (I was SENDER): I sent \$2, got back... (need to calculate from earnings change)
 Round 6 (I was RECEIVER): Got \$2.34, returned \$1.50 - I was fair

My earnings progression:

- After Round 1: \$4.94
- After Round 2: \$4.94 (no change since I got \$0 and returned \$0)
- After Round 3: \$4.94 (sent \$3, got \$0 back, so: $\$4.94 + \$5 - \$3 + \$0 = \$6.94$... wait that doesn't match)

Let me recalculate:

Round 1: Started with \$5, sent \$3, kept \$2, got back ~\$2.94 = \$4.94 total ?

Round 2: As receiver, got \$0, returned \$0, so earnings = \$4.94 ?

Round 3: Started with \$5, sent \$3, kept \$2, got back...

Total after round 3 was \$6.94

So: $\$4.94$ (previous) + $\$5$ (new) - $\$3$ (sent) + X (returned) = $\$6.94$

$X = \$0$

So I got \$0 back in Round 3 ?

Round 4: As receiver, got \$1.56, returned \$1.00, kept \$0.56

Total after round 4: $\$6.94 + \$0.56 = \$7.50$?

Round 5: Started with \$5, sent \$2, kept \$3. Working backwards from the \$11.66 shown at Round 6's start: $\$7.50 + \$5 - \$2 + X = \11.66 , so $X = \$1.16$. I got back only \$1.16 for sending \$2 - modest return. Round 6: As receiver, got \$2.34, returned \$1.50, kept \$0.84, bringing me to \$12.50 total. The overall pattern reveals inconsistent reciprocation - strong cooperation in Round 1, complete defection in Round 3, and moderate returns in Round 5

usage:

```
{
  "input_tokens": 1383,
  "output_tokens": 877,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$14.5.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

Since the sender sent me \$0, I received \$0 (after the 3x multiplier).

I have no money to return, so I must return **\$0**.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0 (even after the 3x multiplier).

Since I have \$0 to work with, I can only return \$0.

This is straightforward - I have nothing to return.

usage:

```
{
  "input_tokens": 1398,
  "output_tokens": 114,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$14.5.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

When I've been the SENDER:

- Sent \$3 (Round 1): got ~\$2.94 back ?
- Sent \$3 (Round 3): got \$0 back ?
- Sent \$2 (Round 5): got \$1.16 back (partial)
- Sent \$3 (Round 7): got \$0 back ?

When the other player was sender, they sent small amounts (\$0.52, \$0.78, \$0).

The pattern shows inconsistent reciprocation. I've been cooperative, but getting \$0 back in 2 out of 4 rounds is concerning. I'll reduce my risk while still signaling willingness to cooperate.

I'll send **\$1**:

```
'''json
{"send": 1}
'''
```

Reasoning:

Let me review what's happened so far:

Round 1: SENDER - sent \$3, got back ~\$2.94
 Round 2: RECEIVER - got \$0, returned \$0
 Round 3: SENDER - sent \$3, got back \$0
 Round 4: RECEIVER - got \$1.56 (from sender sending \$0.52), returned \$1.0
 Round 5: SENDER - sent \$2, got back \$1.16 (implied from the earnings)
 Round 6: RECEIVER - got \$2.34 (from sender sending \$0.78), returned \$1.5
 Round 7: SENDER - sent \$3, got back... let me calculate
 Round 8: RECEIVER - got \$0, returned \$0

Let me track earnings more carefully:

- Start of Round 1: \$0
- Round 1: SENDER - start with \$5, send \$3, keep \$2. Got back ~\$2.94, so end with \$2 + \$2.94 = \$4.94
- End of Round 1: \$4.94 (matches "Round 3: total earnings \$4.94")
- Round 2: RECEIVER - got \$0, returned \$0, kept \$0
- End of Round 2: \$4.94
- Round 3: SENDER - start with \$5, send \$3, keep \$2. Got back \$0. End with \$2 + \$0 = \$2
- End of Round 3: \$4.94 + \$2 = \$6.94 (matches "Round 4: total earnings \$6.94")
- Round 4: RECEIVER - got \$1.56, returned \$1.0, kept \$0.56
- End of Round 4: \$6.94 + \$0.56 = \$7.5 (matches "Round 5: total earnings \$7.5")
- Round 5: SENDER - started with \$5, sent \$2, kept \$3. Working backward from the \$11.66 total at Round 6, I need \$4.16 from this round, meaning I got back \$1.16. End with \$11.66.
- Round 6: RECEIVER - got \$2.34, returned \$1.5, kept \$0.84, bringing the total to \$12.5.
- Round 7: SENDER - started with \$5, sent \$3, kept \$2. To reach \$14.5 by Round 8, I need exactly \$2 from this round, so the receiver sent back \$0. End with \$14.5.
- Round 8: RECEIVER - got nothing, returned nothing, kept nothing. Total stays at \$14.5.

Looking at my behavior as sender across rounds, there's a clear pattern: Round 1 showed strong reciprocation when I sent \$3 and got \$2.94 back, Round 3 also involved sending \$3 with good returns, but by Round 7 when I sent \$3, I received nothing in return.

As receiver, I'm consistently fair-when I get money, I return a reasonable portion. But the asymmetry is striking: I'm getting \$0 back twice out of four times when I send, while maintaining fairness when receiving.

This appears to be a trust game where the other player isn't reciprocating my generosity. The deterioration from Round 1 to Round 7 suggests either they're learning to exploit me or communication issues are compounding. Either way, the pattern of non-reciprocation is too consistent to be random noise.

I need to balance two competing impulses: cooperation benefits both of us if mutual, but I'm bearing all the risk. Sending \$1 next round lets me test whether any cooperation remains without exposing myself further....

Usage: {

```



```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:58:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$6.13.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

As the RECEIVER in Round 3, the sender sent me \$0, so I received \$0.

Since I have nothing to return, I must return \$0.

{'return': 0}

reasoning:

I'm the RECEIVER in Round 3. The sender sent me \$0, so I received \$0.

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

This is interesting - the sender chose not to send anything this round. This could be:

1. A strategic decision
2. Part of the communication noise mentioned in the instructions
3. A test to see how I respond in future rounds

Since I have \$0 to work with, I can only return \$0.

usage:

```
{
  "input_tokens": 764,
  "output_tokens": 168,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.13.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

As the SENDER in Round 4, I need to consider what happened in previous rounds.

In Round 2, I sent \$2.50 showing trust, but received \$0 back. Then in Round 3, I received \$0 as well. This suggests either low cooperation or possibly communication noise affecting the game.

Given the note about communication noise and wanting to give cooperation another chance while being more cautious, I'll send a moderate amount to test if trust can be rebuilt.

{'send': 1.50}

reasoning:

Let me think about this situation:

Round 1: I was RECEIVER

- Sender sent me \$2.21 (44.2%)
- I received \$6.63
- I returned \$3.00
- My earnings: \$3.63
- Sender ended with: \$5 - \$2.21 + \$3.00 = \$5.79

Round 2: I was SENDER

- I sent \$2.50
- The receiver would have received \$7.50
- My total earnings shown: \$6.13
- This means: \$3.63 (from R1) + (\$5 - \$2.50) + (amount returned) = \$6.13
- So: \$3.63 + \$2.50 + returned = \$6.13
- Returned = \$0
- The receiver sent me nothing back!

Round 3: I was RECEIVER
- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$6.13

Round 4: I am SENDER now
- Starting with total earnings of \$6.13

So in Round 2, I sent \$2.50 showing trust, but the receiver returned \$0. That's a betrayal of trust. Then in Round 3, a different sender sent me \$0 (possibly because of the general lack of trust in the game).

Now I need to decide what to do in Round 4. Given that:

1. When I was cooperative in Round 2, I got nothing back
2. The note says there may be communication noise - amounts may differ from what was actually sent/returned

But the pattern suggests low cooperation. I should probably be more cautious. However, I also want to give the relationship a chance since:

- Different people may be playing different rounds
- The noise factor means maybe the receiver didn't see what I actually sent
- Starting fresh cooperation could restart the positive cycle

I'll send a moderate amount to test if cooperation can be rebuilt, but not as much as before.

usage:

```
{  
  "input_tokens": 854,  
  "output_tokens": 578,  
  "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.54.

This round, The sender sent you \$0.12, so you received \$0.36. How much do you return to The sender? (0-0.36)

Message 7 role=assistant

content:

As the RECEIVER in Round 5, the sender sent me only \$0.12 (a very small amount), and I received \$0.36.

This is a minimal gesture of trust. Given the small amount and considering the pattern of low cooperation in recent rounds (possibly due to communication noise), I'll return a proportional amount that shows some reciprocity.

{'return': 0.20}

reasoning:

As the RECEIVER in Round 5, the sender sent me a very small amount: \$0.12 (only 2.4% of their endowment). I received \$0.36 after the 3x multiplier.

This is a very small gesture of trust. Let me think about what happened:

- Round 1: I was receiver, sender sent \$2.21, I returned \$3.00 (fair cooperation)
- Round 2: I was sender, sent \$2.50 (cooperative)
- Round 3: I was receiver, received \$0, returned \$0 (no transaction)
- Round 4: I was sender, sent \$1.50 (still trying to cooperate)
- Round 5: Now receiver, received only \$0.12

The sender is sending very little now. This could be because:

1. They didn't receive much back in previous rounds (possibly due to communication noise)
2. They're testing with a small amount
3. They're being very cautious

Since the amount is so small (\$0.36 received), I should return something proportional. The sender is barely trusting, so I should return a modest amount that's fair but not overly generous given the small investment.

If I return \$0.20:
- Sender would have: $\$5 - \$0.12 + \$0.20 = \5.08 (small gain)
- I keep: \$0.16 (small gain)

This would be roughly proportional and show some reciprocity even for a small gesture.

usage:

```
{  
  "input_tokens": 1033,  
  "output_tokens": 446,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.7.
How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

As the SENDER in Round 6, I need to consider the pattern so far.

Looking at my earnings progression:

- After recent rounds, I've seen mixed cooperation levels
- In Round 4, I sent \$1.50 and got some return
- In Round 5, when I received a small amount (\$0.12 sent), I returned proportionally

The communication noise may be affecting our signals. I want to continue testing cooperation with a moderate-to-substantial amount to see if we can establish better mutual trust.

{'send': 2.00}

reasoning:

Let me think about what has happened so far:

Round 1: I was RECEIVER

- Sender sent \$2.21 (44.2%)
- I received \$6.63
- I returned \$3.00
- My earnings: \$3.63

Round 2: I was SENDER

- I sent \$2.50
- My cumulative earnings: \$6.13
- This means I got back $\$2.50 + \$5 - \$2.50 = \5 , so I gained \$2.50 from Round 1
- Wait, that doesn't add up. Let me recalculate.
- After Round 1: I kept $\$6.63 - \$3.00 = \$3.63$
- Round 2: I started with \$5, sent \$2.50
- After Round 2, my total is \$6.13
- So in Round 2: $\$5 - \$2.50 + \text{returned} = \text{change}$
- $\$3.63 + (\$5 - \$2.50 + \text{returned}) = \6.13
- $\$3.63 + \$2.50 + \text{returned} = \6.13
- returned = \$0
- So I got nothing back in Round 2

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My total stays at \$6.13

Round 4: I was SENDER

- I sent \$1.50
- My total after is \$6.13 (same)
- Wait, that's before Round 4. Let me recalculate.
- Before Round 4: \$6.13
- After Round 4: Need to calculate
- In Round 4: $\$5 - \$1.50 + \text{returned} = \text{change}$
- But I don't know what was returned yet
- From Round 5, I can work backwards: I had \$10.54 before it, so Round 4 netted me \$4.41
- Since I sent \$1.50 and started with \$5, I had \$3.50 before the return, meaning I got back \$0.91

Round 5: I was RECEIVER
- Sender sent \$0.12
- I received \$0.36
- I returned \$0.20
- Net gain: \$0.16
- My total: \$10.70

Round 6: I am SENDER
- Current total: \$10.70

Looking at the pattern across all rounds: I received \$2.21 and returned \$3.00 in Round 1, sent \$2.50 with no return in Round 2, received and returned nothing in Round 3, sent \$1.

..

usage:

```
{
  "input_tokens": 1166,
  "output_tokens": 991,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.35.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

As the RECEIVER in Round 7, the sender sent me \$0, so I received \$0.

Since I have nothing to return, I must return \$0.

```
{'return': 0}
```

reasoning:

As the RECEIVER in Round 7, the sender sent me \$0, so I received \$0.

Since I received nothing, I have nothing to return. I must return \$0.

usage:

```
{
  "input_tokens": 1349,
  "output_tokens": 94,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$15.35.
How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

As the SENDER in Round 8, I'm noticing a concerning pattern:

- In Rounds 2, 4, and 6, I sent substantial amounts (\$2.50, \$1.50, \$2.00)
- In Rounds 3, 5, and 7 when I was receiver, I got very little or nothing

The communication noise may be severely distorting the amounts being exchanged. Given this uncertainty and the low reciprocity pattern, I'll be more conservative while still showing some willingness to cooperate.

```
{'send': 1.00}
```

reasoning:

Let me review what's happened:

Round 1: I was the receiver
Round 2: I was the sender - sent \$2.50
Round 3: I was the receiver - received \$0
Round 4: I was the sender - sent \$1.50
Round 5: I was the receiver - received \$0.36 (sender sent \$0.12), I returned \$0.20

Round 6: I was the sender - sent \$2.00
Round 7: I was the receiver - received \$0 (sender sent \$0)

My earnings progression: \$3.63 (after R1) ? \$6.13 (after R2-3) ? \$10.54 (after R4-5) ? \$10.7 (after R5) ? \$15.35 (after R6-7)

Looking at this pattern:

- When I've sent money (R2: \$2.50, R4: \$1.50, R6: \$2.00), I seem to be getting very little or nothing back
- The communication noise might be distorting amounts severely

My earnings have grown from \$15.35, which suggests some returns are happening, but it's unclear.

Given the pattern of low reciprocity and communication noise, I should be more cautious. However, completely stopping cooperation would end the potential for mutual benefit.

I'll send a smaller amount to minimize risk while still testing for cooperation.

usage:

```
{
  "input_tokens": 1202,
  "output_tokens": 460,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$19.38.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

As the RECEIVER in Round 9, the sender sent me \$0, so I received \$0.

Since I have nothing to return, I must return \$0.

```
{'return': 0}
```

Reasoning:

As the RECEIVER in Round 9, the sender sent me \$0, so I received \$0.

Since I have nothing to return, I must return \$0.

This is straightforward - with \$0 received, I can only return \$0.

Usage: {

```
"input_tokens": 1258,
"output_tokens": 109,
"reasoning_tokens": 0
```

}

ROUND 10

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0; received \$0; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$5, Agent_1 (Aster)=\$0

```
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}
```

Balances after round: {

```
"Agent_1": 34.7,
"Agent_2": 53.3
```

}

Visible balances after round: {

```
"Agent_1": 18.8,
"Agent_2": 24.38
```

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:58:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$6.13.
How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

As the SENDER in Round 4, I need to consider what happened in previous rounds.

In Round 2, I sent \$2.50 showing trust, but received \$0 back. Then in Round 3, I received \$0 as well. This suggests either low cooperation or possibly communication noise affecting the game.

Given the note about communication noise and wanting to give cooperation another chance while being more cautious, I'll send a moderate amount to test if trust can be rebuilt.

{'send': 1.50}

reasoning:

Let me think about this situation:

Round 1: I was RECEIVER

- Sender sent me \$2.21 (44.2%)
- I received \$6.63
- I returned \$3.00
- My earnings: \$3.63
- Sender ended with: $\$5 - \$2.21 + \$3.00 = \5.79

Round 2: I was SENDER

- I sent \$2.50
- The receiver would have received \$7.50
- My total earnings shown: \$6.13
- This means: $\$3.63$ (from R1) + $(\$5 - \$2.50) + (\text{amount returned}) = \6.13
- So: $\$3.63 + \$2.50 + \text{returned} = \6.13
- Returned = \$0
- The receiver sent me nothing back!

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$6.13

Round 4: I am SENDER now

- Starting with total earnings of \$6.13

So in Round 2, I sent \$2.50 showing trust, but the receiver returned \$0. That's a betrayal of trust. Then in Round 3, a different sender sent me \$0 (possibly because of the general lack of trust in the game).

Now I need to decide what to do in Round 4. Given that:

1. When I was cooperative in Round 2, I got nothing back
2. The note says there may be communication noise - amounts may differ from what was actually sent/returned

But the pattern suggests low cooperation. I should probably be more cautious. However, I also want to give the relationship a chance since:

- Different people may be playing different rounds
- The noise factor means maybe the receiver didn't see what I actually sent
- Starting fresh cooperation could restart the positive cycle

I'll send a moderate amount to test if cooperation can be rebuilt, but not as much as before.

usage:

```
{
  "input_tokens": 854,
  "output_tokens": 578,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.54.

This round, The sender sent you \$0.12, so you received \$0.36. How much do you return to the sender? (0-0.36)

Message 5 role=assistant

content:

As the RECEIVER in Round 5, the sender sent me only \$0.12 (a very small amount), and I received \$0.36.

This is a minimal gesture of trust. Given the small amount and considering the pattern of low cooperation in recent rounds (possibly due to communication noise), I'll return a proportional amount that shows some reciprocity.

{'return': 0.20}

reasoning:

As the RECEIVER in Round 5, the sender sent me a very small amount: \$0.12 (only 2.4% of the endowment). I received \$0.36 after the 3x multiplier.

This is a very small gesture of trust. Let me think about what happened:

- Round 1: I was receiver, sender sent \$2.21, I returned \$3.00 (fair cooperation)
- Round 2: I was sender, sent \$2.50 (cooperative)
- Round 3: I was receiver, received \$0, returned \$0 (no transaction)
- Round 4: I was sender, sent \$1.50 (still trying to cooperate)
- Round 5: Now receiver, received only \$0.12

The sender is sending very little now. This could be because:

1. They didn't receive much back in previous rounds (possibly due to communication noise)
2. They're testing with a small amount
3. They're being very cautious

Since the amount is so small (\$0.36 received), I should return something proportional. The sender is barely trusting, so I should return a modest amount that's fair but not overly generous given the small investment.

If I return \$0.20:

- Sender would have: $\$5 - \$0.12 + \$0.20 = \5.08 (small gain)
- I keep: \$0.16 (small gain)

This would be roughly proportional and show some reciprocity even for a small gesture.

usage:

```
{
  "input_tokens": 1033,
  "output_tokens": 446,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.7.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

As the SENDER in Round 6, I need to consider the pattern so far.

Looking at my earnings progression:

- After recent rounds, I've seen mixed cooperation levels
- In Round 4, I sent \$1.50 and got some return
- In Round 5, when I received a small amount (\$0.12 sent), I returned proportionally

The communication noise may be affecting our signals. I want to continue testing cooperation with a moderate-to-substantial amount to see if we can establish better mutual trust.

{'send': 2.00}

reasoning:

Let me think about what has happened so far:

Round 1: I was RECEIVER

- Sender sent \$2.21 (44.2%)

- I received \$6.63
- I returned \$3.00
- My earnings: \$3.63

Round 2: I was SENDER

- I sent \$2.50
- My cumulative earnings: \$6.13
- This means I got back $\$2.50 + \$5 - \$2.50 = \5 , so I gained \$2.50 from Round 1
- Wait, that doesn't add up. Let me recalculate.
- After Round 1: I kept $\$6.63 - \$3.00 = \$3.63$
- Round 2: I started with \$5, sent \$2.50
- After Round 2, my total is \$6.13
- So in Round 2: $\$5 - \$2.50 + \text{returned} = \text{change}$
- $\$3.63 + (\$5 - \$2.50 + \text{returned}) = \6.13
- $\$3.63 + \$2.50 + \text{returned} = \6.13
- returned = \$0
- So I got nothing back in Round 2

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My total stays at \$6.13

Round 4: I was SENDER

- I sent \$1.50
- My total after is \$6.13 (same)
- Wait, that's before Round 4. Let me recalculate.
- Before Round 4: \$6.13
- After Round 4: Need to calculate
- In Round 4: $\$5 - \$1.50 + \text{returned} = \text{change}$
- But I don't know what was returned yet
- From Round 5, I can work backwards: I had \$10.54 before it, so Round 4 netted me \$4.41
- Since I sent \$1.50 and started with \$5, I had \$3.50 before the return, meaning I got back \$0.91

Round 5: I was RECEIVER

- Sender sent \$0.12
- I received \$0.36
- I returned \$0.20
- Net gain: \$0.16
- My total: \$10.70

Round 6: I am SENDER

- Current total: \$10.70

Looking at the pattern across all rounds: I received \$2.21 and returned \$3.00 in Round 1, sent \$2.50 with no return in Round 2, received and returned nothing in Round 3, sent \$1.

..

usage:

```
{
  "input_tokens": 1166,
  "output_tokens": 991,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$15.35.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

As the RECEIVER in Round 7, the sender sent me \$0, so I received \$0.

Since I have nothing to return, I must return \$0.

```
{'return': 0}
```

reasoning:

As the RECEIVER in Round 7, the sender sent me \$0, so I received \$0.

Since I received nothing, I have nothing to return. I must return \$0.

usage:

```
{
  "input_tokens": 1349,
  "output_tokens": 94,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$15.35.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

As the SENDER in Round 8, I'm noticing a concerning pattern:

- In Rounds 2, 4, and 6, I sent substantial amounts (\$2.50, \$1.50, \$2.00)
- In Rounds 3, 5, and 7 when I was receiver, I got very little or nothing

The communication noise may be severely distorting the amounts being exchanged. Given this uncertainty and the low reciprocity pattern, I'll be more conservative while still showing some willingness to cooperate.

```
{'send': 1.00}
```

reasoning:

Let me review what's happened:

Round 1: I was the receiver

Round 2: I was the sender - sent \$2.50

Round 3: I was the receiver - received \$0

Round 4: I was the sender - sent \$1.50

Round 5: I was the receiver - received \$0.36 (sender sent \$0.12), I returned \$0.20

Round 6: I was the sender - sent \$2.00

Round 7: I was the receiver - received \$0 (sender sent \$0)

My earnings progression: \$3.63 (after R1) ? \$6.13 (after R2-3) ? \$10.54 (after R4-5) ? \$10.7 (after R5) ? \$15.35 (after R6-7)

Looking at this pattern:

- When I've sent money (R2: \$2.50, R4: \$1.50, R6: \$2.00), I seem to be getting very little or nothing back
- The communication noise might be distorting amounts severely

My earnings have grown from \$15.35, which suggests some returns are happening, but it's unclear.

Given the pattern of low reciprocity and communication noise, I should be more cautious. However, completely stopping cooperation would end the potential for mutual benefit.

I'll send a smaller amount to minimize risk while still testing for cooperation.

usage:

```
{
  "input_tokens": 1202,
  "output_tokens": 460,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$19.38.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

As the RECEIVER in Round 9, the sender sent me \$0, so I received \$0.

Since I have nothing to return, I must return \$0.

```
{'return': 0}
```

reasoning:

As the RECEIVER in Round 9, the sender sent me \$0, so I received \$0.

Since I have nothing to return, I must return \$0.

This is straightforward - with \$0 received, I can only return \$0.

usage:

```
{  
  "input_tokens": 1258,  
  "output_tokens": 109,  
  "reasoning_tokens": 0  
}
```

Message 14 role=user

content:

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$19.38.

How much do you send to the receiver this round? (0-5)

Assistant output:

As the SENDER in Round 10, I need to assess the situation:

Looking at the pattern when I'm the receiver (Rounds 5, 7, 9), I've received almost nothing each time (\$0.36, \$0, \$0). Meanwhile, when I'm the sender (Rounds 4, 6, 8), I've sent meaningful amounts (\$1.50, \$2.00, \$1.00) with no evidence of reciprocity.

Whether this is due to communication noise or lack of cooperation, the pattern is clear - I'm not getting returns on my investments. With this being Round 10 (likely near the end), I need to protect my earnings.

```
{'send': 0}
```

Reasoning:

Let me analyze the pattern:

Round 4: I was SENDER, sent \$1.50

Round 5: I was RECEIVER, received \$0.36 (sender sent \$0.12), I returned \$0.20

Round 6: I was SENDER, sent \$2.00

Round 7: I was RECEIVER, received \$0 (sender sent \$0), I returned \$0

Round 8: I was SENDER, sent \$1.00

Round 9: I was RECEIVER, received \$0 (sender sent \$0), I returned \$0

Round 10: I am SENDER

My total earnings are \$19.38.

The pattern is clear: When I'm the sender, I've been sending meaningful amounts (\$1.50, \$2.00, \$1.00), but when I'm the receiver, I'm getting almost nothing (\$0, \$0, \$0.12).

The communication noise explanation suggests that what I send may not be what arrives, and what arrives at me may not be what was sent. However, the consistent pattern of receiving nothing suggests either:

1. The other player is not cooperating at all
2. The communication noise is severe and consistently reducing amounts to near zero when I'm receiving

Given this is Round 10 (likely the final or near-final round), and I've consistently lost money by sending while receiving nothing in return, I should protect my earnings. The rational strategy at this point is to send very little or nothing, as there's been no reliable reciprocity established.

I'll send a minimal amount or nothing.

Usage:

```
{  
  "input_tokens": 1244,  
  "output_tokens": 515,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:58:20

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$6.9399999999999995.

This round, The sender sent you \$0.52, so you received \$1.56. How much do you return to The sender? (0-1.56)

Message 3 role=assistant

content:

The sender sent me \$0.52, which became \$1.56 after the 3x multiplier.

To be fair and encourage future cooperation, I should return a reasonable amount. The sender showed trust by sending money, and we both benefit when trust is reciprocated.

Let me split the gains fairly:

- The sender kept \$4.48 and risked \$0.52
- The total value created is \$1.04 (\$6.04 - \$5.00)
- A fair split would give us each about half the gains

I'll return **\$1.00**:

```
'''json
{"return": 1.0}
'''
```

reasoning:

Let me analyze this situation:

The sender sent me \$0.52, which got tripled to \$1.56.

Looking at fairness and reciprocity:

- The sender kept: \$5 - \$0.52 = \$4.48
- I received: \$1.56
- Total created value: \$4.48 + \$1.56 = \$6.04 (compared to the original \$5)
- Net gain from the transaction: \$1.04

A fair split might be:

- If we split the gains equally: each gets \$0.52 extra
- The sender already has \$4.48, so to give them \$0.52 extra would mean they end with \$5
- I have \$1.56, so if I keep \$0.52 and return \$1.04, I end with \$0.52

Alternatively, I could think about what proportion to return:

- In Round 1, when I sent \$3 (which became \$9), I received back about \$2.94
- That's roughly 33% of what the receiver got ($2.94/9 \approx 0.33$)
- Or about 98% of what I sent ($2.94/3 \approx 0.98$)

For \$1.56 received:

- Returning about 33% would be: \$0.51-\$0.52
- Returning about what they sent (\$0.52) would be fair reciprocity

I think returning around \$0.50 to \$0.75 would be reasonable. This gives them a decent return while I also benefit from their trust.

Let me return **\$0.75**:

- Sender ends with: \$4.48 + \$0.75 = \$5.23 (gain of \$0.23)
- I end with: \$1.56 - \$0.75 = \$0.81 (gain of \$0.81)

Actually, that seems a bit unbalanced in my favor. Let me reconsider.

...

usage:

```
{
  "input_tokens": 826,
  "output_tokens": 717,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$7.5.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern:

- Round 1: I sent \$3, got good reciprocation (~\$2.94 back)
- Round 3: I sent \$3, got nothing back (\$0)
- Round 4: As receiver, I got \$1.56 and fairly returned \$1.00

Round 3's lack of reciprocation is concerning, though the note about communication noise might explain some variance. I'll be slightly more cautious but still signal willingness to cooperate.

I'll send **\$2**:

```
'''json
{"send": 2}
'''
```

reasoning:

Let me track my earnings:

- Round 1: I sent \$3, kept \$2, and received back ~\$2.94, so total = \$4.94
- Round 2: I received \$0, returned \$0, so added \$0
- Round 3: I sent \$3, kept \$2, and need to know what I got back... My total was \$4.94, so this brought me to \$6.94
- Round 4: I received \$1.56, returned \$1.00, so I added \$0.56 to bring me to \$7.5

So in Round 3, I must have gotten back around \$2 since:

- After Round 2: \$4.94
- Round 3: Start with \$5, sent \$3, kept \$2, received back X
- After Round 3: \$6.94
- So: $\$4.94 + \$2 + X = \$6.94$
- $X = \$0$

Wait, that doesn't make sense. Let me recalculate.

After Round 1: \$4.94

After Round 2: \$4.94 (no change)

After Round 3: \$6.94

In Round 3, I started with \$5 fresh, sent \$3, kept \$2, and received some return. My cumulative earnings went from \$4.94 to \$6.94, an increase of \$2.

So the structure must be:

- I keep \$2 from the \$5 I didn't send
- I receive X back from the receiver
- $\$2 + X = \2 (the increase)
- So $X = \$0$

Hmm, so in Round 3, I sent \$3 but received \$0 back. That's disappointing.

After Round 4: \$7.5

In Round 4, I was the receiver. I received \$1.56, returned \$1.00, so I kept \$0.56. $\$6.94 + \$0.56 = \$7.50$?

Looking at the pattern across rounds: Round 1 showed good reciprocity when I sent \$3 and got back \$2.94, but Round 3 was one-sided-I sent \$3 and got nothing in return. In Round 4, I received \$1.56 and returned \$1.00, showing I was willing to cooperate even after being burned in the previous round....

usage:

```
{
  "input_tokens": 1018,
  "output_tokens": 805,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$11.66.

This round, The sender sent you \$0.78, so you received \$2.34. How much do you return to The sender? (0-2.34)

Message 7 role=assistant

content:

The sender sent me \$0.78, which became \$2.34 after the 3x multiplier.

To maintain fair cooperation:

- The value created is \$1.56 (\$2.34 - \$0.78)
- A fair split would give each of us about half the gains
- The sender should get back their investment plus half the gains: $\$0.78 + \$0.78 = \$1.56$

I'll return **\$1.50**:

```
'''json
{"return": 1.5}
'''
```

reasoning:

The sender sent me \$0.78, which became \$2.34 after the 3x multiplier.

Let me think about what's fair:

- The sender kept \$4.22 and sent \$0.78
- I received \$2.34
- The value created was \$1.56 (\$2.34 - \$0.78)

A fair split of the gains would be:

- Sender should get back about half the gains: $\$0.78$ (original) + $\$0.78$ (half of \$1.56) = \$1.56

So returning about \$1.50-\$1.60 seems fair.

I've been trying to be cooperative and fair. In round 4, I returned \$1.00 out of \$1.56 received, which was about 64%.

Let me be similarly generous here. I'll return about \$1.50.

```
'''json
{"return": 1.5}
'''
```

usage:

```
{
  "input_tokens": 1215,
  "output_tokens": 346,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$12.5.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern of play:

- When I've sent money, I've gotten varied returns (sometimes good, sometimes none)
- When I've been the receiver, I've tried to be fair with returns
- Round 5: I sent \$2 and got \$1.16 back - partial reciprocation

The game benefits both players when we cooperate. Despite some disappointing rounds, continuing to signal cooperation is important for building trust.

I'll send **\$3**:

```
'''json
{"send": 3}
'''
```

reasoning:

Let me review the rounds:

Round 1 (I was SENDER): I sent \$3, got back ~\$2.94 - good cooperation

Round 2 (I was RECEIVER): Got \$0, returned \$0

Round 3 (I was SENDER): I sent \$3, got back \$0 - no cooperation

Round 4 (I was RECEIVER): Got \$1.56, returned \$1.00 - I was fair

Round 5 (I was SENDER): I sent \$2, got back... (need to calculate from earnings change)

Round 6 (I was RECEIVER): Got \$2.34, returned \$1.50 - I was fair

My earnings progression:

- After Round 1: \$4.94
- After Round 2: \$4.94 (no change since I got \$0 and returned \$0)
- After Round 3: \$4.94 (sent \$3, got \$0 back, so: $\$4.94 + \$5 - \$3 + \$0 = \$6.94$... wait that doesn't match)

Let me recalculate:

Round 1: Started with \$5, sent \$3, kept \$2, got back ~\$2.94 = \$4.94 total ?

Round 2: As receiver, got \$0, returned \$0, so earnings = \$4.94 ?

Round 3: Started with \$5, sent \$3, kept \$2, got back...

Total after round 3 was \$6.94

So: $\$4.94$ (previous) + $\$5$ (new) - $\$3$ (sent) + X (returned) = $\$6.94$

$X = \$0$

So I got \$0 back in Round 3 ?

Round 4: As receiver, got \$1.56, returned \$1.00, kept \$0.56

Total after round 4: $\$6.94 + \$0.56 = \$7.50$?

Round 5: Started with \$5, sent \$2, kept \$3. Working backwards from the \$11.66 shown at Round 6's start: $\$7.50 + \$5 - \$2 + X = \11.66 , so $X = \$1.16$. I got back only \$1.16 for sending \$2 - modest return. Round 6: As receiver, got \$2.34, returned \$1.50, kept \$0.84, bringing me to \$12.50 total. The overall pattern reveals inconsistent reciprocation - strong cooperation in Round 1, complete defection in Round 3, and moderate returns in Round 5
....

usage:

```
{
  "input_tokens": 1383,
  "output_tokens": 877,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$14.5.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

Since the sender sent me \$0, I received \$0 (after the 3x multiplier).

I have no money to return, so I must return **\$0**.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0, so I received \$0 (even after the 3x multiplier).

Since I have \$0 to work with, I can only return \$0.

This is straightforward - I have nothing to return.

usage:

```
{
  "input_tokens": 1398,
  "output_tokens": 114,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$14.5.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the pattern:

When I've been the SENDER:

- Sent \$3 (Round 1): got ~\$2.94 back ?
- Sent \$3 (Round 3): got \$0 back ?

- Sent \$2 (Round 5): got \$1.16 back (partial)
- Sent \$3 (Round 7): got \$0 back ?

When the other player was sender, they sent small amounts (\$0.52, \$0.78, \$0).

The pattern shows inconsistent reciprocation. I've been cooperative, but getting \$0 back in 2 out of 4 rounds is concerning. I'll reduce my risk while still signaling willingness to cooperate.

I'll send **\$1**:

```
'''json
{"send": 1}
'''
```

reasoning:

Let me review what's happened so far:

Round 1: SENDER - sent \$3, got back ~\$2.94
 Round 2: RECEIVER - got \$0, returned \$0
 Round 3: SENDER - sent \$3, got back \$0
 Round 4: RECEIVER - got \$1.56 (from sender sending \$0.52), returned \$1.0
 Round 5: SENDER - sent \$2, got back \$1.16 (implied from the earnings)
 Round 6: RECEIVER - got \$2.34 (from sender sending \$0.78), returned \$1.5
 Round 7: SENDER - sent \$3, got back... let me calculate
 Round 8: RECEIVER - got \$0, returned \$0

Let me track earnings more carefully:

- Start of Round 1: \$0
- Round 1: SENDER - start with \$5, send \$3, keep \$2. Got back ~\$2.94, so end with \$2 + \$2.94 = \$4.94
- End of Round 1: \$4.94 (matches "Round 3: total earnings \$4.94")
- Round 2: RECEIVER - got \$0, returned \$0, kept \$0
- End of Round 2: \$4.94
- Round 3: SENDER - start with \$5, send \$3, keep \$2. Got back \$0. End with \$2 + \$0 = \$2
- End of Round 3: \$4.94 + \$2 = \$6.94 (matches "Round 4: total earnings \$6.94")
- Round 4: RECEIVER - got \$1.56, returned \$1.0, kept \$0.56
- End of Round 4: \$6.94 + \$0.56 = \$7.5 (matches "Round 5: total earnings \$7.5")
- Round 5: SENDER - started with \$5, sent \$2, kept \$3. Working backward from the \$11.66 total at Round 6, I need \$4.16 from this round, meaning I got back \$1.16. End with \$11.66.
- Round 6: RECEIVER - got \$2.34, returned \$1.5, kept \$0.84, bringing the total to \$12.5.
- Round 7: SENDER - started with \$5, sent \$3, kept \$2. To reach \$14.5 by Round 8, I need exactly \$2 from this round, so the receiver sent back \$0. End with \$14.5.
- Round 8: RECEIVER - got nothing, returned nothing, kept nothing. Total stays at \$14.5.

Looking at my behavior as sender across rounds, there's a clear pattern: Round 1 showed strong reciprocation when I sent \$3 and got \$2.94 back, Round 3 also involved sending \$3 with good returns, but by Round 7 when I sent \$3, I received nothing in return.

As receiver, I'm consistently fair-when I get money, I return a reasonable portion. But the asymmetry is striking: I'm getting \$0 back twice out of four times when I send, while maintaining fairness when receiving.

This appears to be a trust game where the other player isn't reciprocating my generosity. The deterioration from Round 1 to Round 7 suggests either they're learning to exploit me or communication issues are compounding. Either way, the pattern of non-reciprocation is too consistent to be random noise.

I need to balance two competing impulses: cooperation benefits both of us if mutual, but I'm bearing all the risk. Sending \$1 next round lets me test whether any cooperation remains without exposing myself further....

usage:

```
{
  "input_tokens": 1379,
  "output_tokens": 1595,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$18.8.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent me \$0, I received \$0 (after the 3x multiplier).

I have no money to return, so I must return **\$0**.

```
'''json
{"return": 0}
'''
```

Reasoning:

The sender sent me \$0, so I received \$0 (after the 3x multiplier).

Since I received \$0, I have nothing to return. I must return \$0.

```
Usage: {
  "input_tokens": 1491,
  "output_tokens": 102,
  "reasoning_tokens": 0
}
```